

Software Development

Student Edition

Presented By
LK LearnKey®

Software Development Project Workbook

First Edition

LearnKey creates signature multimedia courseware. LearnKey provides expert instruction for popular computer software, technical certifications, and application development with dynamic video-based courseware and effective learning management systems. For a complete list of courses, visit <https://www.learnkey.com>.

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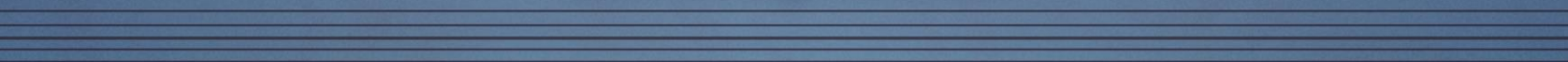
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Introduction

Software
Development

A decorative graphic consisting of several thin, light blue lines that resemble circuit traces or a stylized border. These lines run horizontally across the lower half of the image, with some segments featuring small circles or branching patterns, giving it a technical or digital feel.

Best Practices Using LearnKey's Online Training

LearnKey offers video-based training solutions that are flexible enough to accommodate private students and educational facilities and organizations.

Our course content is presented by top experts in their respective fields and provides clear and comprehensive information. The full line of LearnKey products has been extensively reviewed to meet superior standards of quality. Our course content has also been endorsed by organizations such as Certiport, CompTIA®, Cisco, and Microsoft. However, it is the testimonials given by countless satisfied customers that truly set us apart as leaders in the information training world.

LearnKey experts are highly qualified professionals who offer years of job and project experience in their subjects. Each expert has been certified at the highest level available for their field of expertise. This expertise provides the student with the knowledge necessary to obtain top-level certifications in their chosen field.

Our accomplished instructors have a rich understanding of the content they present. Effective teaching encompasses presenting the basic principles of a subject and understanding and appreciating organization, real-world application, and links to other related disciplines. Each instructor represents the collective wisdom of their field and within our industry.

Our Instructional Technology

Each course is independently created based on the manufacturer's standard objectives for which the course was developed.

We ensure that the subject matter is up-to-date and relevant. We examine the needs of each student and create training that is both interesting and effective. LearnKey training provides auditory, visual, and kinesthetic learning materials to fit diverse learning styles.

Course Training Model

The course training model allows students to undergo basic training, building upon primary knowledge and concepts to more advanced application and implementation. In this method, students will use the following toolset:

Pre-assessment: The pre-assessment is used to determine the student's prior knowledge of the subject matter. It will also identify a student's strengths and weaknesses, allowing them to focus on the specific subject matter they need to improve the most. Students should not necessarily expect a passing score on the pre-assessment as it is a test of prior knowledge.

Video training sessions: Each training course is divided into sessions or domains and lessons with topics and subtopics. LearnKey recommends incorporating all available external resources into your training, such as student workbooks, glossaries, course support files, and additional customized instructional material. These resources are located in the folder icon at the top of the page.

Exercise labs: Labs are interactive activities that simulate situations presented in the training videos. Step-by-step instructions and live demonstrations are provided.

Post-assessment: The post-assessment is used to determine the student's knowledge gained from interacting with the training. In taking the post-assessment, students should not consult the training or any other materials. A passing score is 80 percent or higher. If the individual does not pass the post-assessment the first time, LearnKey recommends incorporating external resources, such as the workbook and additional customized instructional material.

Workbook: The workbook has various activities, such as fill-in-the-blank worksheets, short answer questions, practice exam questions, and group and individual projects that allow the student to study and apply concepts presented in the training videos.

Using This Workbook

This project workbook contains practice projects and exercises to reinforce the knowledge you have gained through the video portion of the **Software Development** course. The purpose of this workbook is twofold. First, get you further prepared to pass the Software Development Fundamentals exam, and second, to get you job-ready skills and increase your employability in the area of software development.

The projects within this workbook follow the order of the video portion of this course. To save your answers in this workbook, you must first download a copy to your computer. You will not be able to save your answers in the web version. You can complete the workbook exercises as you go through each section of the course, complete several of these at the end of each domain, or complete them after viewing the entire course. The key is to go through these projects to strengthen your knowledge in this area.

Each project is based upon a specific video (or videos) in the course and specific test objectives. The materials you will need for this course include:

- LearnKey's **Software Development** courseware.
- Visual Studio Code or any development tool that can run C#, Java, Python, HTML, CSS, and JavaScript code.
- The course project files. All applicable project files are located in the support area where you downloaded this workbook.

For Teachers

LearnKey is proud to provide extra support to instructors upon request. For your benefit as an instructor, we also provide an instructor support .zip file containing answer keys, completed versions of the workbook project files, and other teacher resources. This .zip file is available within your learning platform's admin portal.

Notes

- Extra teacher notes, when applicable, are in the Project Details box within each exercise.
- Exam objectives are aligned with the course objectives listed in each project, and project file names correspond with these numbers.
- Short answers may vary but should be similar to those provided in this workbook.
- The Finished folder in each domain has reference versions of each project. These can help you grade projects.
- Teachers may consider asking students to add their initials, student ID, or other personal identifiers at the end of each saved project.
- Refer to your course representatives for further support.

We value your feedback about our courses. If you have any questions, comments, or concerns, please let us know by visiting <https://about.learnkey.com>.

Skills Assessment

Instructions: Rate your skills on the following tasks from 1-5 (1 being needs improvement, 5 being excellent).

Skills	1	2	3	4	5
Describe computer storage and data types.	☐	☐	☐	☐	☐
Construct and analyze algorithms and flowcharts to solve programming problems.	☐	☐	☐	☐	☐
Incorporate error handling into applications or modules.	☐	☐	☐	☐	☐
Construct and analyze code based on functional programming patterns.	☐	☐	☐	☐	☐
Describe software development lifecycle (SDLC) management.	☐	☐	☐	☐	☐
Interpret application specifications.	☐	☐	☐	☐	☐
Construct and analyze code that uses algorithms and data structures.	☐	☐	☐	☐	☐
Describe the purpose of version control systems.	☐	☐	☐	☐	☐
Describe secure coding concepts.	☐	☐	☐	☐	☐
Construct, analyze, and use classes.	☐	☐	☐	☐	☐
Construct and analyze code that uses inheritance.	☐	☐	☐	☐	☐
Construct and analyze code that uses polymorphism.	☐	☐	☐	☐	☐
Construct and analyze web applications.	☐	☐	☐	☐	☐
Describe and configure web hosting.	☐	☐	☐	☐	☐
Describe and configure web services.	☐	☐	☐	☐	☐
Describe and identify architectural patterns.	☐	☐	☐	☐	☐
Design and normalize a database.	☐	☐	☐	☐	☐
Construct, analyze, and optimize ANSI SQL queries.	☐	☐	☐	☐	☐
Manage transactions.	☐	☐	☐	☐	☐
Describe database access methods.	☐	☐	☐	☐	☐

Skills	1	2	3	4	5
Describe types of NoSQL databases.	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

Software Development Video Times

Domain 1	Video Time
Computer Storage and Data Types	00:08:31
Algorithms and Flowcharts	00:15:22
Error Handling	00:07:47
Functional Programming Patterns	00:06:12
Total Time	00:37:52

Domain 2	Video Time
SDLC Management	00:07:18
Interpret Application Specifications	00:01:59
Code With Algorithms and Data Structures	00:12:01
Version Control Systems	00:08:38
Secure Coding Concepts	00:07:03
Total Time	00:36:59

Domain 3	Video Time
Classes	00:12:57
Inheritance	00:05:41
Polymorphism	00:05:53
Total Time	00:24:31

Domain 4	Video Time
Construct and Analyze Web Applications	00:14:27
Configure Web Hosting	00:04:12
Configure Web Services	01:05:50
Architectural Patterns	00:02:30
Total Time	01:26:59

Domain 5	Video Time
Design and Normalize a Database	00:18:41
SQL Queries	00:17:34
Manage Transactions	00:10:15
Database Access Methods	00:04:01
NoSQL Databases	00:02:22
Total Time	00:48:52

Domain 1

Lesson 1

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 1 Lesson 1, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. Data stored in _____ use either a specific or varying amount of memory in storage. [How Things Are Stored]
2. Coders need to ensure that once memory is no longer needed, the memory space used for data or a task is _____. [Memory Stacks and Heaps]
3. In C#, integers use _____ bytes of memory. [Memory Size Requirements]
4. In languages such as C#, garbage collection should automatically happen when memory runs _____ or an object exceeds its allocated memory space. [Garbage Collection]

Reading and Storing Code

One aspect of writing efficient code is knowing how a computer stores programs and instructions in computer memory. Data stored in variables use either a specific or varying amount of memory in storage.

C#, Java, Python, and other programming languages are high-level languages, meaning writing code in a readable form. Then when run, it is compiled into a lower-level state and then executed.

Purpose

Upon completing this project, you will better understand how to use code within Visual Studio Code.

Steps for Completion

1. C# code is one of several programming languages that, when compiled, is morphed into _____.
2. Open Visual Studio Code or another similar program.
3. Search for the latest step-by-step instructions for setting up C# projects within Visual Studio.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Computer Storage and Data Types

Subtopic: How Things Are Stored

Objectives covered

1 Core Programming Concepts

1.1 Describe computer storage and data types

1.1.1 How a computer stores programs and instructions in computer memory

Memory Stacks and Heaps

One aspect of programming is to use as little memory as possible to perform a task. For that reason, coders should ensure that once memory is no longer needed, the memory space used for data or a task is deallocated. Memory stacks and heaps can affect how users write code as both are memory spaces in a computer's RAM.

Purpose

Upon completing this project, you will better understand the difference between memory stacks and heaps related to memory usage within a program.

Steps for Completion

1. Stacks use static or fixed memory allocation and include items such as variables. Determine whether each statement about variables is true or false. Use T for true and F for false.
 - a. _____ A variable stores a value that can be used later in an app.
 - b. _____ Variables can use stacks because their data type does not define how much memory is needed.
2. When the amount of memory needed is not defined, one should use heaps. Determine whether each statement about heaps is true or false. Use T for true and F for false.
 - a. _____ The amount of memory needed in a heap needs to be adjustable or dynamic.
 - b. _____ Heaps are faster than stacks, so coders should use heaps when possible.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Computer Storage and Data Types

Subtopic: Memory Stacks and Heaps

Objectives covered

1 Core Programming Concepts

1.1 Describe computer storage and data types

1.1.2 Memory stacks and heaps

Memory Size Requirements

One aspect of memory and storage is the number of bytes each data type utilizes within programs. For both C# and Java, integers, which are whole numbers, use four bytes of memory. Doubles, which allow for decimals and can be up to 15 digits in length, use eight bytes of memory. Strings use two bytes per character, as C# uses Unicode, allowing for symbols and foreign language characters to be included in text.

Purpose

Upon completing this project, you will better understand memory size requirements.

Steps for Completion

1. View the code below. How many bytes does the phrase, hello, everyone!, use?

a. _____

```

4  class Program
5  {
6      static void Main(string[] args)
7      {
8          Console.WriteLine("hello, everyone!");
9          // integers use four bytes of memory
10         // doubles use eight bytes of memory
11         // Strings use two bytes per character
12     }
13 }
```

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Computer Storage and Data Types

Subtopic: Memory Size Requirements

Objectives covered

1 Core Programming Concepts

1.1 Describe computer storage and data types

1.1.3 Memory size requirements for various data storage types, numeric data, and textual data

Garbage Collection

Garbage collection is a form of automatic memory management. Variables and collections are allocated memory, and that memory space is usually freed up once a variable or collection is no longer needed. This process is like a garbage collector's set schedule for emptying specific dumpsters. In languages such as C#, garbage collection should automatically happen when memory runs low, or an object exceeds its allocated memory space.

Purpose

Upon completing this project, you will better understand how to run the garbage collection feature.

Steps for Completion

1. When might a coder want to invoke a garbage collection?
 - a. _____

 _____.
2. Open the **Program.cs** file from your Domain 1 Student folder.
3. Below the existing code, add a line of code: **GC.Collect();**
4. On the next line, add: **Console.WriteLine("Memory used: {0}", GC.GetTotalMemory(true));**
5. Run the code and view the memory in use.

Project Details

Project file

Program.cs

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Computer Storage and Data Types

Subtopic: Garbage Collection

Objectives covered

1 Core Programming Concepts

1.1 Describe computer storage and data types

1.1.4 Garbage collection

Domain 1

Lesson 2

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 1 Lesson 2, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. There are three types of decision structures: _____, _____, and conditional loops.
[Decision Structures]
2. The IF decision structure requires action based on whether a condition is _____. [If Statements]
3. An efficient way to code for multiple conditions is to use a _____ statement. [Multiple Decision Structures]
4. As part of the software development process, a flowchart can and should be used to show the _____ flow of an app. [Flowcharts]
5. Part of a _____ contains decision logic which can be expanded into a decision table. [Decision Tables]
6. The concept of evaluating expressions is the act of figuring out a _____ with multiple operands.
[Evaluating Expressions]
7. A for each loop runs as many times as needed to cover an entire _____ of items. [For Loops]
8. A while loop will not run if the condition set for it is never _____. [While Loops]
9. If you only need to see the contents of a conditional loop run strictly if a condition is true, use a _____ loop. [Do...While Loops]
10. Recursion occurs when a subroutine or function calls _____ within code. [Recursion]

Decision Structures

Decision structures are used in almost every programming language, with some variation. There are essentially three types of decision structures: conditions, loops, and conditional loops.

Purpose

Upon completing this project, you will better understand decision structures and if statements.

Steps for Completion

1. In a(n) _____, a condition is presented. If the condition is true, a block of code runs. If not, either nothing happens, a different condition is checked, or a different code block runs.
2. A _____ runs a code loop a specific number of times according to what is set in the first section of the code.
3. A _____ is a conditional loop that runs if a condition is true. The loop will run at least once.
4. View the if statement on lines 13 and 14.

```

6      class Program
7      {
8          static void Main(string[] args)
9          {
10             double disc;
11             string customerStatus = "Silver";
12
13             if (customerStatus == "Gold")
14                 disc=.1;
15         }
16     }
    
```

5. According to the if statement, if the customerStatus is _____, the customer gets a 10% discount on an order.
6. According to the code, what is set to occur if the customer status is not Gold? Explain.

a. _____

Project Details

Project file

N/A

Estimated completion time

5-10 minutes

Video reference

Domain 1

Topic: Algorithms and Flowcharts

Subtopic: Decision Structures; If Statements

Objectives covered

1 Core Programming Concepts

1.2 Construct and analyze algorithms and flowcharts to solve programming problems

1.2.1 Decision structures used in all computer programming languages

1.2.2 If decision structures

Multiple Decision Structures

Most decision structures are not used for just one condition and one decision. Many decision structures have multiple conditions to be evaluated, with a decision based on any of those conditions being true.

Using if, else if, and else statements allow one to set multiple conditions and take actions based on those conditions. One crucial component of an if statement is that whichever condition is met first will dictate the action taken, and then the rest of the if statement is ignored.

In statements with numeric conditions, the order in which the conditions are presented matters. Logic order is important in a conditional statement because the input amounts often affect one another.

With most programming languages, users can have several else if conditions as part of an if statement. However, the more conditions there are, the tougher they are to manage. A more efficient way to code multiple conditions is to use a switch statement, including multiple cases, each representing a condition.

Purpose

Upon completing this project, you will better understand decision structures and their uses.

Steps for Completion

1. View the if else statement on lines 13-15.

```

8      static void Main(string[] args)
9      {
10         double disc;
11         string customerStatus = "Silver";
12
13         if (customerStatus == "Gold")
14             disc=.1;
15         else disc=.05;
16     }
```

2. Explain what the if else statement means.

- a. _____

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 1

Topic: Algorithms and Flowcharts

Subtopic: Multiple Decision Structures

Objectives covered

1 Core Programming Concepts

1.2 Construct and analyze algorithms and flowcharts to solve programming problems

1.2.3 Multiple decision structures, such as if...else and switch

3. View the if else statement on lines 13-17.

```
13     if (customerStatus == "Gold")
14         disc=.1;
15     else if (customerStatus == "Silver")
16         disc=.05;
17     else disc=0;
18 }
```

4. Explain what the updated if else statement means.

a. _____

_____.

5. The switch statement shown has three cases, and within each case, a discount is represented based on customer status. Why are there break statements at the end of each case?

a. _____

_____.

6. The default value in this switch statement is _____.

7. The _____ keyword sends the statement result to where it needs to go for further processing.

```
case "Gold":
    disc = .1;
    break;
case "Silver":
    disc = .05;
    break;
case "Bronze":
    disc = .02;
    break;
default:
    disc= 0;
    break;
return;
```

Flowcharts and Decision Tables

As part of the software development process, a flowchart can and should be used to show the logic flow of an app. A program is easier to code when its logic has already been defined. Flowcharts use shapes to represent specific happenings within a program. If necessary, users can build their flowchart in pieces and then connect them to build the overall logic flow for an app.

Part of a flowchart is decision logic, and, sometimes, that part should be expanded into a decision table. Decision tables help coders map out variables leading to a decision and show how code needs to be written for these decisions. The logic within a decision table should be incorporated into a flowchart and then subsequently into programming code once one gets to that stage in developing an app. Displayed is a matrix that can be used as a decision table.

Promo	Gold	Silver	Bronze
None	10%	5%	2%
Hot	20%	15%	12%
Fall	15%	10%	7%

Purpose

Upon completing this project, you will better understand how and when to use flowcharts and decision tables.

Steps for Completion

1. Match the shapes to the items they represent within a flowchart.

- | | |
|-------------------|-------------------|
| A. Rounded shapes | D. Parallelograms |
| B. Diamonds | E. Rectangles |
| C. Arrows | |

- _____ The order in which logic flows through an app
- _____ Processes
- _____ Inputs and outputs
- _____ The start and end of an algorithm
- _____ Decision-making operations

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Algorithms and Flowcharts

Subtopic: Flowcharts; Decision Tables

Objectives covered

1 Core Programming Concepts

1.2 Construct and analyze algorithms and flowcharts to solve programming problems

1.2.4 Reading and constructing flowcharts

1.2.5 Decision tables

Evaluating Expressions

Evaluating expressions is the act of performing a calculation with multiple operands. When evaluating expressions, one should know and understand the order of operations, which occurs in this order: parentheses, exponents, multiplication, division, addition, and subtraction. Understanding the order of operations is important because having an incorrect expression will lead to an incorrect result and create logic problems within the code.

Purpose

Upon completing this project, you will be able to solve expressions using the order of operations.

Steps for Completion

- Many people use the expression “Please Excuse My Dear Aunt Sally” to remember the order of operations. Use the table to create your own phrase to help you remember this order.

P	Parentheses	_____
E	Exponents	_____
M	Multiplication	_____
D	Division	_____
A	Addition	_____
S	Subtraction	_____

- Use the order of operations to solve the expressions.

- $10+3*2=$ _____
- $(10+3)*2=$ _____
- $4*(2+7)=$ _____
- $6+3-4*11=$ _____

Project Details

Project file

N/A

Estimated completion time

5-10 minutes

Video reference

Domain 1

Topic: Algorithms and Flowcharts

Subtopic: Evaluating Expressions

Objectives covered

1 Core Programming Concepts

1.2 Construct and analyze algorithms and flowcharts to solve programming problems

1.2.6 Evaluating expressions

For Loops

Loops run a block of code either a set number of times or a varying number of times based on a true or false condition. A for loop runs a block of code a set number of times. A for statement contains three pieces: a variable set to a starting point, a variable that determines the condition in which the loop will keep running, and code that states the increments in which a loop runs.

A for each loop runs as many times as needed to run through an entire collection of items.

Purpose

Upon completing this project, you will better understand how and when to use for loops.

Steps for Completion

1. View the for statement on lines 10-13.

```

7      {
8          static void Main(string[] args)
9          {
10             for(int i = 1; i<= 3; i++)
11             {
12                 Console.WriteLine("Funday!");
13             }

```

2. Explain what the for loop is supposed to accomplish when run.

a. _____

3. View the for each string on lines 10-13.

```

8          static void Main(string[] args)
9          {
10             Console.WriteLine("Locations:");
11             string[] cities = { "Aurora","Centennial","Louisville" };
12             foreach (string s in cities)
13             {
14                 Console.WriteLine(s);
15             }

```

4. Explain what the for each loop is supposed to accomplish when run.

a. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Algorithms and Flowcharts

Subtopic: For Loops

Objectives covered

1 Core Programming Concepts

1.2 Construct and analyze algorithms and flowcharts to solve programming problems

1.2.7 For loops

While and Do...While Loops

Not all loops need to run a fixed number of times. Many times, a loop should only run while a condition is true or false. One should use a while loop in this situation. However, a while loop will not run if its condition is never true.

Those who need to see something in a conditional loop run at least once should use a do...while loop. If they only need to see the contents of a conditional loop run strictly if a condition is true, they should use a while loop.

Purpose

Upon completing this project, you will better understand while loops.

Steps for Completion

1. View the while loop on lines 10-14.

```

8      static void Main(string[] args)
9      {
10     int x = 1;
11     while (x <= 5)
12     {
13         Console.WriteLine("You have {0} orders of toy cars", x);
14         x++;
15     }
16 }

```

2. Explain what the while loop is supposed to accomplish when run.

a. _____

3. View the do...while loop on lines 10-16.

```

8      static void Main(string[] args)
9      {
10     int x = 1;
11     do
12     {
13         Console.WriteLine("You have {0} orders of toy cars", x);
14         x++;
15     }
16     while (x <= 5);
17 }

```

4. The while statement on line 16 indicates that while x is less than or equal to _____, the loop should keep running.

5. Why use a do...while loop instead of a while loop?

a. _____

Project Details

Project file

N/A

Estimated completion time

5-10 minutes

Video reference

Domain 1

Topic: Algorithms and Flowcharts

Subtopic: While Loops; Do...While Loops

Objectives covered

1 Core Programming Concepts

1.2 Construct and analyze algorithms and flowcharts to solve programming problems

1.2.8 While loops

1.2.9 Do...while loops

Recursion

Recursion is another programming concept. Recursion occurs when a subroutine or function calls itself within code.

Purpose

Upon completing this project, you will better understand recursion.

Steps for Completion

1. View the code example, which consists of a trivia game. On line 15, the Console.ReadLine method is used to get input from a person. The person must try to guess the capital of Colorado. If their answer is correct, they will win. If their answer is not correct, what will occur?

a. _____

 _____.

```

10      string result = question();
11      Console.WriteLine(result);
12
13      string question() {
14          Console.WriteLine("What is the capital of Colorado?");
15          string answer = Console.ReadLine();
16
17          if (answer == "Denver")
18              {Console.WriteLine("You win!");}
19          else question();
20          return "Game over";}
21      }
```

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Algorithms and Flowcharts

Subtopic: Recursion

Objectives covered

1 Core Programming Concepts

1.2 Construct and analyze algorithms and flowcharts to solve programming problems

1.2.10 Recursion

Domain 1

Lesson 3

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 1 Lesson 3, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. One goal of software development should be to make sure that errors within an app do not cause the app to _____. [Structured Exception Handling]
2. Unit testing tests a single unit of _____ to ensure it functions as it should. [Unit Testing]
3. One can generate an _____ at any time by throwing an exception. [Throwing Exceptions]
4. Knowing how to read a stack can help users pinpoint _____ in code. [Reading Stacks]
5. Another way to employ defensive coding is to limit what can be entered in a _____. [Defensive Coding]
6. There are two types of scoping: static and _____. [Scoping in Exception Handling]
7. Events are _____, such as a mouse selection, a field on a form changing, or a window opening. [Event]
8. Delegates protect _____ that should not be exposed in code to others. [Delegate]
9. A good practice is to double-check with others before officially _____ code that is not being used. [Promises]
10. Making code asynchronous allows multiple tasks to be _____ at once. [Synchronous vs. Asynchronous]
11. Immutability is the act of having a _____ that cannot be changed once code is compiled and an app is running. [Immutability]

Structured Exception Handling

One goal of software development should be to make sure that errors within an app do not cause the app to crash. One way of accomplishing this is through structured exception handling. Structure exception handling is running code to see if it works and using a different block of code if the first does not work properly.

Purpose

Upon completing this project, you will better understand structured exception handling.

Steps for Completion

1. View the try-catch block on lines 12-16.

```

8      static void Main(string[] args)
9      {
10     int x= 5;
11     int y= 0;
12     try
13     {
14         Console.WriteLine(x/y);
15     }
16     catch (Exception ex)
17     {
18         Console.WriteLine(ex.Message);
19     }

```

2. Explain what the try block is supposed to accomplish when run.

a. _____

3. When added, a _____ block will display regardless of whether the try block worked or the catch block ran because the try block did not work.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Error Handling

Subtopic: Structured Exception Handling

Objectives covered

1 Core Programming Concepts

1.3 Incorporate error handling into applications or modules

1.3.1 Structured exception handling (try-catch-finally)

Unit Testing and Exceptions

Errors can be found during a period of testing known as unit testing. Unit testing tests a single unit of code to ensure it functions as it should. When testing a unit of code, it should be tested for every possibility, especially if the code is conditional.

While structured exception handling can be accomplished through try-catch blocks, coders may want more control over when an error message appears. One can generate an error message at any point in their code by throwing an exception.

Purpose

Upon completing this project, you will better understand the different types of unit testing and when to throw an exception.

Steps for Completion

1. _____ testing is testing done with a certain expected outcome.
2. _____ testing is typically done by users without knowledge of any expected outcome.
3. View the throw exception on line 14.

```

9      {
10     int x= 5;
11     int y= 0;
12     if (y==0)
13     {
14         throw new InvalidOperationException("Cannot divide by zero");
15     }
16     else
17     {
18         Console.WriteLine(x/y);
19     }
20 }
```

4. Explain what the throw exception is supposed to accomplish when run.

a. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Error Handling

Subtopic: Unit Testing; Throwing Exceptions

Objectives covered

1 Core Programming Concepts

1.3 Incorporate error handling into applications or modules

1.3.2 Unit testing

1.3.3 Throwing exceptions

Reading Stacks and Defensive Coding

When an app does cause an error, it is important to find out where the error occurred. Designers can accomplish this task by reading a stack of errors that shows when running an app that contains an error. Knowing how to read a stack can help users pinpoint errors in code.

One way to avoid errors when programming is to employ defensive coding or coding to ensure software functions under normal and unforeseen circumstances. This type of coding can be accomplished through structured exception handling, throwing exceptions, and limiting what can be entered in a data field. Limiting what can be entered in a data field can involve limiting values or controlling which data type is used for input.

Purpose

Upon completing this project, you will better understand how to read a stack and employ defensive coding.

Steps for Completion

1. View the stack output.

```
Determining projects to restore...
All projects are up-to-date for restore.
C:\Users\JasonManibog\OneDrive - LearnKey, Inc\Desktop\Software Development Fundamentals\Domain1\Program.cs(10,17): error CS1002
: ; expected [C:\Users\JasonManibog\OneDrive - LearnKey, Inc\Desktop\Software Development Fundamentals\Domain1\Domain1.csproj]
The terminal process "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -Command dotnet build /property:GenerateFullPath
s=true /consoleloggerparameters:NoSummary" terminated with exit code: 1.
```

2. On what line and which character contains the error?
 - a. _____
3. View the code snippet. This code asks users to input the number of toy cars they want. Users must enter a whole number, so the amount is set to an integer, not a double, so no decimals are entered.

```
8      static void Main(string[] args)
9      {
10         try {
11             Console.WriteLine("How many cars do you want?");
12             int amount = Convert.ToInt32(Console.ReadLine());
13             Console.WriteLine("Your order total will be {0} dollars", amount*16);
14         }
15
16         catch (Exception ex)
17         {
18             Console.WriteLine(ex.Message);
19         }
```

4. When the code is run, and users enter something other than an integer, the code will throw an error and stop. What would you do to prevent this from happening?
 - a. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Error Handling

Subtopic: Reading Stacks;
Defensive Coding

Objectives covered

1 Core Programming Concepts

1.3 Incorporate error handling into applications or modules

1.3.4 Reading the stack

1.3.5 Defensive coding

Scoping in Exception Handling

The last part of error handling is scoping in exception handling. There are two types of scoping: static and dynamic. Static scoping refers to reviewing the code itself to spot problems in the code before running it. Dynamic scoping involves running code and determining which aspects run and which do not.

Purpose

Upon completing this project, you will better understand scoping in exception handling.

Steps for Completion

1. View the code below. In this example, nothing stands out as a potential error. However, when it is run and something other than an integer is entered, an error occurs.
2. Which type of scoping is this known as?
 - a. _____

```
Console.WriteLine("How many cars do you want?");
int amount = Convert.ToInt32(Console.ReadLine());
Console.WriteLine("Your order total will be {0} dollars", amount*16);
```

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Error Handling

Subtopic: Scoping in Exception Handling

Objectives covered

1 Core Programming Concepts

1.3 Incorporate error handling into applications or modules

1.3.6 Understand scoping in exception handling

Events and Delegates

One type of functional programming pattern is events. Events are happenings, such as a mouse selection, a field on a form changing, or a window opening. Events trigger actions, otherwise known as methods.

Another functional programming pattern is the use of delegates. Delegates protect functions that should not be exposed in code to others as they can be used, for example, to call a method within a class, which is a template for an object.

Purpose

Upon completing this project, you will better understand events and delegates.

Steps for Completion

1. Open the **141-event.html** file from your Domain 1 Student folder.
2. Preview the webpage.
3. Select the Show the Sale message button.
4. Which events occurred in the process outlined above?
 - a. _____
5. A delegate allows one to tap into a _____ without referencing a class itself, which can be useful if it is marked as private.

Project Details

Project file

141-event.html

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Functional Programming Patterns

Subtopic: Event; Delegate

Objectives covered

1 Core Programming Concepts

1.4 Construct and analyze code based on functional programming patterns

1.4.1 Event

1.4.2 Delegate

Promises

Promises are conceptual programming patterns. The concept of promises basically states that every function in code should have a defined purpose, or one should consider not having the function in the code.

Purpose

Upon completing this project, you will be able to review code to ensure that each line serves a purpose.

Steps for Completion

1. Open the **143-promises.html** file from your Domain 1 Student folder.
2. Review the code to ensure that each line has a function.
3. Comment out the class code starting on line 15 as it is not used anywhere else in the code.
 - a. On line 14, type: **/*** at the beginning of the line
 - b. On line 20, type: ***/** at the end of the line
4. Within the comment, add the note: **This code is not being used and should be deleted.**

Project Details

Project file

143-promises.html

Estimated completion time

5-10 minutes

Video reference

Domain 1

Topic: Functional Programming Patterns

Subtopic: Promises

Objectives covered

1 Core Programming Concepts

1.4 Construct and analyze code based on functional programming patterns

1.4.3 Promises

Synchronous vs. Asynchronous

In programming, most tasks work synchronously, meaning that one task completes and then the next one occurs. This sequence works well within a linear process, but it may not be the best solution for a situation where people do not want to wait for an entire interface to load before interacting with it. One solution is to take a block of code and make it asynchronous. Making code asynchronous allows multiple tasks to be processed at once.

Purpose

Upon completing this project, you will be able to make a task asynchronous.

Steps for Completion

1. Open the **144-synchronous vs. asynchronous.txt** file from your Domain 1 Student folder.
2. The showWarning method on line 18 needs to be complete before the calculation on line 22 can occur. Make these tasks asynchronous.
 - a. Add the keyword **async** in front of Task on line 12.
 - b. Add the keyword **await** after return on line 22.
3. Asynchronous JavaScript and XML (AJAX) allows for JavaScript and XML processes to run _____.
4. XMLHttpRequest Object (XHR) is used to request and receive data from a _____.

Project Details

Project file

144-synchronous vs. asynchronous.txt

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Functional Programming Patterns

Subtopic: Synchronous vs. Asynchronous

Objectives covered

1 Core Programming Concepts

1.4 Construct and analyze code based on functional programming patterns

1.4.4 Synchronous vs. asynchronous programming (AJAX and XHR)

Immutability

The final programming pattern in this section is immutability. Immutability is the act of having a variable that cannot be changed once code is compiled and an app is running.

Purpose

Upon completing this project, you will better understand immutability.

Steps for Completion

1. _____ are variables that cannot be changed once they are assigned a value. That is an example of immutability.
2. Classes are blueprints for _____. Any private property in a class is considered immutable.
3. By default, strings are _____, meaning that if you change a string value in code, the string variable is deleted and replaced with a new one.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 1

Topic: Functional Programming
Patterns

Subtopic: Immutability

Objectives covered

1 Core Programming Concepts

1.4 Construct and analyze code
based on functional programming
patterns

1.4.5 Immutability

Domain 2

Lesson 1

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 2 Lesson 1, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. Requirement analysis is the process of detailing _____ requirements for an app. [Requirement Analysis]
2. In the Design Process phase, the two main designs are the technical design and the _____ design. [Planning and Design]
3. Most projects need a _____ to build and oversee any data storage areas built for an app. [Implementation]
4. In the Testing stage of the software development life cycle (SDLC), three major types of tests need to occur: unit tests, functional tests, and _____ tests. [Testing]
5. Deployment often involves creating an installation package for an app, such as an MSI file or an _____. [Deployment]
6. The Maintenance phase of Release Management manages _____ identified as people use the software. [Maintenance]
7. Agile, similar to traditional software development, has a cycle of steps that include _____, Design, Develop, Test, and _____. [Agile Concepts]
8. Often code should be broken down into _____ before being developed. [Read Application Specifications]
9. C# can be used for standalone apps, both console or command-based, and with _____. [Select Appropriate Types and Components]

Requirement Analysis

Coders must be aware of the software development life cycle (SDLC). All phases are important to understand as one needs to know the goals for the software they are developing before writing code. Once a software application need is identified, a project manager is chosen. At that point, one can start the first SDLC phase: requirements gathering and analysis, also known as requirements analysis.

Requirement analysis is the process of detailing business requirements for an application. Functional requirements are the what-the-app-must-do requirements, while non-functional requirements include needs such as reliability and scalability.

Purpose

Upon completing this project, you will better understand the elements of a requirement analysis.

Steps for Completion

1. Open the **211-requirements.docx** file from your Domain 2 Student folder
2. Review the requirement analysis for a toy store app.
3. List at least two of the functional requirements included in the sample.
 - a. _____
 - _____
4. List at least one of the non-functional requirements included in the sample.
 - a. _____

Project Details

Project file

211-requirements.docx

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: SDLC Management

Subtopic: Requirement Analysis

Objectives covered

2 Software Development Principles

2.1 Describe Software Development Life Cycle (SDLC) management

2.1.1 Requirement analysis

Planning, Design, and Implementation

The next phase of the SDLC is planning and design, also known as the design process. In this phase, the two main designs created are the technical design and the user interface design. These designs matter because they show a development team the logic for and desired interface of an app.

After the design process comes implementation, also known as development. Implementation should not be confused with deployment, which is the actual distribution of an app. Developers should write code in the implementation process, but there are other, equally important, tasks and members of a development team.

Purpose

Upon completing this project, you will better understand the planning and implementation elements of a project.

Steps for Completion

1. A technical design of an app can be created in the form of a _____.
2. A user interface design is a _____ of what an app should look like when completed.
3. Most projects need a database administrator (DBA) to build and oversee _____ being built for an app.
4. _____ are needed to write documentation, training manuals, and help files associated with an app.
5. Content _____ are needed to add actual data to an app.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: SDLC Management

Subtopic: Planning and Design;
Implementation

Objectives covered

2 Software Development Principles

2.1 Describe Software Development
Life Cycle (SDLC) management

2.1.2 Planning and design

2.1.3 Implementation

Testing and Deployment

Once an app is built, it needs to be thoroughly tested before it can be deployed. In the testing stage of the SDLC, three major types of tests need to occur: security tests, functional tests, and unit tests.

Security tests ensure that an app is only accessible to those who need it and that permissions are granted properly to those who need them. Functional tests ensure an app works the way it is supposed to according to the business requirements set forth. Unit tests make sure a unit of code works.

Once testing is complete and an app works the way it is meant to, it is ready to be deployed. Deployment often involves creating an installation package for the app, such as an MSI file or an executable, except in the case of web apps. Development and database teams should be ready for any last-minute fixes to an app as it is being deployed.

Purpose

Upon completing this project, you will be able to test code and know when it is ready for deployment.

Steps for Completion

1. Open the **Program.cs** file from your Domain 2 Student folder.
2. Run a unit test on the code.
3. Did the code run properly, without any errors?
 - a. _____
4. A web app can be moved to a _____ web server and be deployed when ready.
5. When deploying an app, a team of _____ should be identified and used to test and make sure the app is working as it should as it is being deployed. This process is called user acceptance testing (UAT).

Project Details

Project file

Program.cs

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: SDLC Management

Subtopic: Testing; Deployment

Objectives covered

2 Software Development Principles

2.1 Describe Software Development Life Cycle (SDLC) management

2.1.4 Testing

2.1.5 Deployment

Maintenance

The final step in the SDLC is maintenance. Technically, both deployment and maintenance are part of release management. The maintenance part of release management manages defects identified as people use the released software. In this maintenance phase, software enhancements and change requests are also identified and prioritized.

Purpose

Upon completing this project, you will better understand the maintenance process.

Steps for Completion

1. Open the **216-Maintenance.docx** file from your Domain 2 Student folder.
2. Review the typical list of bugs, enhancements, and change requests one may have for an app.
3. Within the maintenance process, a team needs to determine the _____ of any enhancement or change requests.
4. Some change requests are implemented as soon as possible, with proper testing, and some are implemented upon the next _____ change of an app.

Project Details

Project file

216-Maintenance.docx

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: SDLC Management

Subtopic: Maintenance

Objectives covered

2 Software Development Principles

2.1 Describe Software Development Life Cycle (SDLC) management

2.1.6 Maintenance

Agile Concepts

One popular method of development is Agile. Agile, similar to traditional software development, has a cycle of steps that include Requirements, Design, Develop, Test, and Deploy, which is pretty similar to the SDLC. The differences lie in the way app development takes place.

The Agile approach is more iterative than the SDLC and is well-suited for fluid projects, with changing requirements as the actual development occurs. With the ability to make and publish changes quickly, this development method works well if it is well-managed.

Purpose

Upon completing this project, you will better understand Agile development.

Steps for Completion

1. List the four key principles of Agile development.

- a. _____

- b. _____

- c. _____

- d. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: SDLC Management

Subtopic: Agile Concepts

Objectives covered

2 Software Development Principles

2.1 Describe Software Development

Life Cycle (SDLC) management

2.1.7 Agile Concepts

Read Application Specifications

Developers rarely go from defining program requirements straight into writing code. Often, code should be broken down into prototypes before being developed.

Logic defined for an app can be broken down into pseudocode, which is basically an English language version of code. Having pseudocode makes it easier to determine what the actual code should look like once the languages for an app have been established.

Purpose

Upon completing this project, you will better understand requirements and pseudocode.

Steps for Completion

1. View the requirements outlined in the example below.
2. Why would someone choose to write code requirements in paragraph form?
 - a. _____
 - _____
 - _____
 - _____
 - _____

3. View the pseudocode for the displayed requirements.

Requirement: Customers with a gold status get 10% off their orders, silver 5%, and bronze 2%.

Pseudocode

```
If a customer is gold, order discount is 10%
or if a customer is silver, 5%
or if a customer is bronze, 2%
or else no discount
```

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Interpret Application Specifications

Subtopic: Read Application Specifications

Objectives covered

2 Software Development Principles

2.2 Interpret application specifications

2.2.1 Reading application specifications and translating them into prototypes and code

Select Appropriate App Types

Building an app involves knowing what platforms said app needs to run on. Developers should ask: does it need to run as a standalone app? Is it going to be a web app? Does it need to run on mobile devices? All these factors must be considered to determine what languages to use for the app.

Most web apps use HTML, CSS, and JavaScript, and they often use a server-side language such as C#. C# can also be used for standalone apps, both console or command-based, and with one or more graphical user interfaces.

Purpose

Upon completing this project, you will better understand the different application development tools available for app creation.

Steps for Completion

1. C# is used heavily on _____ platforms, while Java is more used in open-source situations.
2. The internal and external _____ will drive app needs.
3. _____ is a popular language that is considered easier to write code for than C# or Java.
4. Python is an _____ language, not a compiled language like C# or Java, and thus runs slower than does C# or Java.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Interpret Application Specifications

Subtopic: Select Appropriate Types and Components

Objectives covered

2 Software Development Principles

2.2 Interpret application specifications

2.2.2 Selecting appropriate application type and components

Domain 2

Lesson 2

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 2 Lesson 2, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. An array can store a group of values in a single _____. [Arrays]
2. An example of a _____ scenario is having boxes of items where the last one put on the shelf is the first one distributed to a customer. [Stacks]
3. Queues can consist of different _____ types. [Queues]
4. A linked list is both _____ and can be adjusted in multiple ways. [Linked Lists]
5. Dictionaries allow people to define custom _____ for items without being limited to fields in a database. [Dictionaries]
6. In a _____ sort, data is broken up into smaller sets, the sets are sorted, and then the sets are merged. [Sorting Algorithms]
7. A linear search is a _____ search through a list of items until a list item is found. [Searching Algorithms]
8. Arrays are highly structured as they are fixed in _____, so they typically perform better than linked lists or dictionaries. [Performance Implications]
9. Queues are _____ lists, making them well-suited for lists where the oldest items need to be removed first. [Choosing the Best Data Structure]

Arrays

Developers must understand data structures. One such structure is arrays. An array can store a group of values in a single variable. Most arrays are fixed in nature, meaning their values can change, but the number of values typically does not change. If the array is set to be dynamic, the number of values in the array can change.

Purpose

Upon completing this project, you will better understand arrays.

Steps for Completion

1. View the code sample below.
2. On which line/s is the array?
 - a. _____
3. What does the array contain?
 - a. _____

```

8      static void Main(string[] args)
9      {
10         string[] locations = {"Centennial", "Denver", "Louisville", "Vail"};
11         Console.WriteLine(locations[0]);
12         // locations[3]="Lakewood";
13         // Console.WriteLine(locations[3]);
14     }
  
```

4. Most arrays are zero-based, meaning that value 0 is the _____ value.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Code With Algorithms and Data Structures

Subtopic: Arrays

Objectives covered

2 Software Development Principles

2.3 Construct and analyze code that uses algorithms and data structures

2.3.1 Arrays

Stacks

Stacks are structured lists that operate on a last-in-first-out (LIFO) basis. An example of LIFO is having boxes of items where the last one put on the shelf is the first one distributed to a customer. Stacks can have varying data types, such as strings and integers.

Purpose

Upon completing this project, you will better understand the use of stacks in code.

Steps for Completion

1. Open the **Program.cs** file from your Domain 2 Student folder.
2. On line 10, there is a string-based stack declared called _____.
3. On lines 11 through 14, there is a _____ method, which adds items to the stack.
4. On line 18, there is a _____ loop that loops through each item in the stack collection and displays the item.
5. Run the code.
 - a. View the four items that have been pushed to the stack (in the Terminal display).
6. Uncomment the code on line 15.
7. The Pop command displays the _____ element in the stack and then removes it.
8. Run the code.
 - a. Notice that 104 has been popped off the stack.
9. Uncomment the code on line 22.
10. The Peek method shows the _____ item in the stack.
11. Run the code.
 - a. Notice that the results are the same except that item 103 shows twice.

Project Details

Project file

Program.cs

Estimated completion time

5-10 minutes

Video reference

Domain 2

Topic: Code With Algorithms and Data Structures

Subtopic: Stacks, LIFO

Objectives covered

2 Software Development Principles

2.3 Construct and analyze code that uses algorithms and data structures

2.3.2 Stacks

2.3.11 LIFO

Queues

A queue is a type of data structure that uses the first-in-first-out (FIFO) method for managing the data in the queue. Using a queue works well for waiting lists where the oldest items need to come off the list first. Queues can be made up of many different data types.

Purpose

Upon completing this project, you will better understand the use of queues in code.

Steps for Completion

1. Open the **Program.cs** file from your Domain 2 Student folder.
2. On line 10, there is a string-based queue declared called _____.
3. On lines 11 through 14, there is an _____ method.
4. On line 16, there is a _____ loop that loops through and displays each item in the collection.
5. Run the code.
 - a. View the three order numbers listed.
6. Uncomment the code on line 14.
7. The _____ statement removes an item from the queue.
8. Run the code.
 - a. Notice the two remaining orders.
9. Uncomment the code on line 20.
10. The _____ statement displays the first item in the queue.
11. Run the code.
 - a. Notice that order 11112 has become the first order on the list.

Project Details

Project file

Program.cs

Estimated completion time

5-10 minutes

Video reference

Domain 2

Topic: Code With Algorithms and Data Structures

Subtopic: Queues, FIFO

Objectives covered

2 Software Development Principles

2.3 Construct and analyze code that uses algorithms and data structures

2.3.3 Queues

2.3.10 FIFO

Linked Lists and Dictionaries

This course has covered arrays, stacks, and queues. Arrays store a list of values and are generally fixed and not easily expandable. Stacks and queues are expandable, but they utilize a LIFO method and FIFO method of changing values in lists, respectively.

A linked list is expandable and has multiple ways of adjusting the list. Dictionaries allow people to define custom categories for each item and not be limited to fields in a database. Dictionaries offer versatility in data storage as they allow each item to have a varying set of characteristics.

Purpose

Upon completing this project, you will better understand using linked lists and dictionaries in code.

Steps for Completion

1. Open the **Program.cs** file from your Domain 2 Student folder.
2. On line 10, there is a string-based linked list declared called _____.
3. On lines 11 through 14, there is an _____ method, which adds each item as the last one on the list.
4. On line 18, there is a _____ loop that loops through and displays each item in the collection.
5. Run the code.
 - a. View the four colors listed.
6. Uncomment the code on line 16.
7. The _____ method can remove any item from the list.
8. Run the code.
 - a. Notice the green element is removed from the collection.
9. View the JSON dictionary example. Which object has a trunk property?
 - a. _____
10. This code is an example of _____ data with a group of objects with similar characteristics, but not all the fields are the same.

Project Details

Project file

Program.cs

Estimated completion time

5-10 minutes

Video reference

Domain 2

Topic: Code With Algorithms and Data Structures

Subtopic: Linked Lists; Dictionaries

Objectives covered

2 Software Development Principles

2.3 Construct and analyze code that uses algorithms and data structures

2.3.4 Linked Lists

2.3.5 Dictionaries (key-value pairs)

```

1  {
2  "toycars": {
3    "toycar1":{
4      "Name":"Cobra",
5      "Color":"black"
6    },
7    "toycar2":{
8      "Name":"Stingray",
9      "Color":"maroon",
10     "Trunk":"yes",
11     }
12   }
13 }
```

Sorting Algorithms

The next series of algorithms to understand are sort algorithms. Developers should know how to sort an array, the different sort types available, and how they work.

Purpose

Upon completing this project, you will better understand how to use sorting in code.

Steps for Completion

1. Open the **Program.cs** file from your Domain 2 Student folder.
2. On line 10, there is a string-based array declared called _____. The _____ method on the array will put the array in alphabetical order.
3. On line 14, there is a _____ loop that writes out each location.
4. Run the code.
 - a. Notice that the list of cities is in alphabetical order.
5. Uncomment the code on line 12.
6. The _____ method applied to the array will put the array in descending order.
7. Run the code.
 - a. Notice that the items in the array are listed in descending alphabetical order.
8. In a _____ sort, each element is sorted one by one and slotted accordingly in the sort.
9. In a _____ sort, each element is compared to the next element in the list, and if the elements need to be swapped, they are swapped.
10. In a _____ sort, an element becomes the pivot point, and then elements are placed to the left and right of the pivot element accordingly.
11. In a _____ sort, data is broken up into smaller sets, the sets are sorted, and then the sets are merged.

Project Details

Project file

Program.cs

Estimated completion time

5-10 minutes

Video reference

Domain 2

Topic: Code With Algorithms and Data Structures

Subtopic: Sorting Algorithms

Objectives covered

2 Software Development Principles

2.3 Construct and analyze code that uses algorithms and data structures

2.3.6 Sorting algorithms (selection sort, bubble sort, quick sort, and merge sort)

Searching Algorithms

Searching algorithms identify how items are searched for within an array or something like an array. Two types of searches to know about include a linear search, a sequential search through a list of items until a list item is found, and a binary search, a recursive search through identifying a middle element until that element matches what is being searched.

Purpose

Upon completing this project, you will better understand searches that can be used in code.

Steps for Completion

1. View the example of the code search below.
2. What is the searchResult variable set to search for?
 - a. _____
3. To see all the matches for search criteria, one can change Find to _____ and then re-run the code.

```

8      static void Main(string[] args)
9      {
10         string[] locations = {"Centennial", "Denver", "Greeley", "Lakewood"};
11
12         var searchResult = Array.Find(locations, element => element.StartsWith("L"))
13
14         Console.WriteLine(searchResult);
15     }
    
```

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Code With Algorithms and Data Structures

Subtopic: Searching Algorithms

Objectives covered

2 Software Development Principles

2.3 Construct and analyze code that uses algorithms and data structures

2.3.7 Searching algorithms (linear search and binary search)

Performance Implications and Choosing Data Structures

Developers should be able to view the various data structures and how they work from a performance standpoint. Any data structure that is well structured and organized will have better performance overall. However, each data structure can be useful in the right context with the correct application.

Purpose

Upon completing this project, you will be able to evaluate data structures to determine what a project may need best.

Steps for Completion

1. Determine whether each statement about data structures is true or false. Use T for true and F for false.
 - a. _____ Arrays are highly structured as they are fixed in size.
 - b. _____ Arrays typically do not perform better than linked lists or dictionaries.
 - c. _____ Stacks and queues are fast because they are very well-defined.
 - d. _____ Neither stacks nor queues are well-suited for searches because, unlike arrays, they do not really mark elements with a location.
 - e. _____ In C#, an array does not need to be redeclared before its size can be adjusted.
 - f. _____ Queues are last-in-first-out (LIFO) lists, making them well-suited for when the most recent items on a list need to be removed first.
 - g. _____ Stacks are first-in-first-out (FIFO) lists, making them well-suited for lists where the oldest items need to be removed from a list first.
 - h. _____ Linked lists are versatile in that they support items of different data types, and the lists are easy to maneuver, expand, and update.
 - i. _____ Dictionaries are best to use when a semi-structured data set is needed.

Project Details

Project file

N/A

Estimated completion time

5-10 minutes

Video reference

Domain 2

Topic: Code With Algorithms and Data Structures

Subtopic: Performance Implications; Choosing the Best Data Structure

Objectives covered

2 Software Development Principles

2.3 Construct and analyze code that uses algorithms and data structures

2.3.8 Performance implications of various data structures

2.3.9 Choosing the right data structure

Domain 2

Lesson 3

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 2 Lesson 3, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. GitHub is a platform used to host _____ files and offers version control and collaborative services. [GitHub]
2. When one wants to work on a file, one should check out a file to help _____ others from working on the same file. [Check Out]
3. It is always a good idea to add a _____ to any new file version. [Check In]
4. Branching allows multiple developers to have multiple copies of a _____, make changes, and merge changes with other branches. [Branch]
5. A merge starts with a _____ request. [Merge]
6. To work with a _____ from the GitHub website, users need to clone it. [Clone]
7. If someone discovers that a file change was incorrect and needs to be rolled back to a previous commit, they cannot accomplish the task through the _____ version of GitHub. [Rollback]
8. If two people change the same line within the same _____, a conflict will arise when merging changes. [Resolve Conflicts]
9. Digital signatures improve _____ in a file as it goes from one recipient to another. [Encryption, Hashing, and Digital Signatures]
10. _____ can be symmetric or asymmetric. [Keys]
11. Cross-site request forgery (CSRF) is an attack in which someone _____ into a website is forced into executing unwanted actions on the site. [Mitigating CSRF]
12. SQL injection occurs when someone puts SQL code into a _____ field, and the code is processed verbatim. [SQL Injection]
13. Malicious users can hijack a person's _____ actions inside of an IFrame. [Risks of Using IFrames]

GitHub

GitHub is a platform used to host code files. GitHub offers both version control and collaborative services. Understanding GitHub’s capabilities allow developers to manage code files effectively.

Purpose

Upon completing this project, you will better understand GitHub.

Steps for Completion

1. What is version control?
 - a. _____

 _____.
2. What is a repository ready to do once it has been created?
 - a. _____
 _____.
3. After files have been uploaded to a repository, what must be selected to make the files available for use?
 - a. _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Version Control Systems

Subtopic: GitHub

Objectives covered

2 Software Development Principles

2.4 Describe the purpose of version control systems

2.4.1 GitHub

Check Out and Check In

GitHub has features that make collaborative work easier to manage. The ability to check out a file is useful when multiple people have access to the file. After a file has been checked out, it must be checked in for others to work on the file.

Purpose

Upon completing this project, you will better understand checking out and checking in a file in GitHub.

Steps for Completion

1. What is the purpose of checking out a file?

a. _____

_____.

2. What does red indicate when previewing changes in GitHub?

a. _____.

3. What does green indicate when previewing changes in GitHub?

a. _____.

4. What is a good idea to add to a file that one is checking in?

a. _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Version Control Systems

Subtopic: Check Out; Check In

Objectives covered

2 Software Development Principles

2.4 Describe the purpose of version control systems

2.4.3 Check-out

3.4.2 Check-in

Branch

It is possible to create branches in GitHub. The main branch contains all of the code files in a repository. Developers can create copies of the main branch.

Purpose

Upon completing this project, you will better understand branches.

Steps for Completion

1. What is the purpose of branching?

a. _____

_____.

2. When is the main branch formed?

a. _____
_____.

3. Why might one choose to create a new branch?

a. _____
_____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Version Control Systems

Subtopic: Branch

Objectives covered

2 Software Development Principles

2.4 Describe the purpose of version control systems

2.4.5 Branch

Merge and Clone

Merging files can help eliminate confusion when multiple versions of a file exist. For some developers, cloning a repository may be necessary. Understanding how to merge files and clone repositories can help developers have the files they need and keep files organized.

Purpose

Upon completing this project, you will better understand merging and cloning within GitHub.

Steps for Completion

1. What does a merge do?
 - a. _____
_____.
2. What begins a merge?
 - a. _____.
3. Why might one need to clone a repository?
 - a. _____
_____.
4. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ People can approve their own changes when merging a file.
 - b. _____ The purpose of branches is to have one person working on multiple files.
 - c. _____ To view the files within a cloned repository, one should select Show in Explorer.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Version Control Systems

Subtopic: Merge; Clone

Objectives covered

2 Software Development Principles

2.4 Describe the purpose of version control systems

2.4.4 Merge

2.4.7 Clone

Rollback and Resolving Conflicts

Occasionally, problems may arise when storing files on GitHub. A file may need to be rolled back, or a conflict may be discovered within a file. Knowing how to find and manage these issues helps developers ensure that their files stored on GitHub are correct. In addition, GitHub supports an overall code review process to ensure the correct code is being used in apps built from code stored in GitHub repositories.

Purpose

Upon completing this project, you will better understand how to roll back a file and resolve conflicts in a file.

Steps for Completion

1. Why might a file need to be rolled back to a previous commit?
 - a. _____

 _____.
2. Why would a conflict arise when merging changes?
 - a. _____

 _____.
3. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ A file can only be rolled back in the desktop version of GitHub.
 - b. _____ To see if there are any conflicts in a file, one needs to push the changes to the server first.
4. Where are code reviews initiated from within GitHub?
 - a. _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Version Control Systems

Subtopic: Rollback; Resolve Conflicts; Code Review

Objectives covered

2 Software Development Principles

2.4 Describe the purpose of version control systems

2.4.6 Rollback

2.4.8 Resolving conflicts

2.4.9 Code review

Encryption, Hashing, and Digital Signatures

Data often needs to be protected due to regulations or the risk of harm if data is exposed. Encryption, hashing, and digital signatures provide different ways of making data secure.

Purpose

Upon completing this project, you will better understand encryption, hashing, and digital signatures.

Steps for Completion

1. Match the term to its description.

Hashing	Digital signatures	Encryption
---------	--------------------	------------

- a. _____ A mechanism used to improve integrity in a file as it goes from one recipient to another
 - b. _____ The scrambling of data so that it is undecipherable without a key
 - c. _____ The scrambling of data, but it is one-way
2. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ Credit card numbers should always be encrypted.
 - b. _____ When using hashes, if the two hashes are different, then the data is considered secure.
 - c. _____ Digital signatures use encryption.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Secure Coding Concepts

Subtopic: Encryption, Hashing, and Digital Signatures

Objectives covered

2 Software Development Principles

2.5 Describe secure coding concepts

2.5.1 Encryption, hashing, and digital signatures

Keys

Sometimes, regulations, such as those found for the Payment Card Industry Data Security Standard (PCI DSS), mandate that certain data be encrypted. One can generate encryption keys to protect sensitive data. Encryption can be symmetric or asymmetric.

Purpose

Upon completing this project, you will better understand encryption keys.

Steps for Completion

1. What is symmetric encryption?
 - a. _____

2. What is asymmetric encryption?
 - a. _____

3. What is the Advanced Encryption Standard (AES) used to create?
 - a. _____
4. What is Rivest-Shamir-Adleman (RSA) used to create?
 - a. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Secure Coding Concepts

Subtopic: Keys

Objectives covered

2 Software Development Principles

2.5 Describe secure coding concepts

2.5.2 Public, private, and shared keys

Mitigating CSRF

One possible security problem within applications is cross-site request forgery (CSRF). While CSRF poses a risk to users, anti-CSRF tokens can be used to prevent CSRF.

Purpose

Upon completing this project, you will better understand CSRF.

Steps for Completion

1. What is CSRF?
 - a. _____

 _____.
2. When does CSRF occur?
 - a. _____

 _____.
3. What is an anti-CSRF token?
 - a. _____

 _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 2

Topic: Secure Coding Concepts

Subtopic: Mitigating CSRF

Objectives covered

2 Software Development Principles

2.5 Describe secure coding concepts

2.5.3 Mitigating cross-site request forgery (CSRF)

SQL Injection and IFrames

SQL injections are a risk to applications, specifically those containing form fields. Though SQL injections pose a risk, there are multiple ways to prevent them. IFrames, while useful, are also vulnerable in several ways. Understanding SQL injections and the risks of IFrames can help developers be better prepared to manage risks.

Purpose

Upon completing this project, you will better understand SQL injection and IFrames.

Steps for Completion

1. What is a SQL injection?

a. _____

 _____.

2. What are the risks associated with SQL injection?

a. _____
 _____.

3. What are two common ways to prevent SQL injection?

a. _____
 _____.

4. What do IFrames do?

a. _____.

5. What are the three ways in which IFrames make a webpage vulnerable?

a. _____

 _____.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 2

Topic: Secure Coding Concepts

Subtopic: SQL Injection; Risks of Using IFrames

Objectives covered

2 Software Development Principles

2.5 Describe secure coding concepts

2.5.4 SQL injection

2.5.5 Risks of using IFrames

Domain 3

Lesson 1

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 3 Lesson 1, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. An object is an _____ that can be used in code. [Properties, Methods, Events, Fields, and Constructors]
2. Classes contain properties for an object and often contain methods that use the object to drive _____. [Create Classes]
3. Once a class is created, the next goal is usually to create an _____. [Use Classes in Code]
4. A private property is not exposed to users within an app except through a _____ property, allowing for control. [Access Modifiers]
5. Declaring an _____ based on a class is known as an instantiation of a class. [Instantiation]
6. A static class can be copied without _____ an instance of that class as an object. [Static vs. Instance]
7. Encapsulation is the act of shielding pieces of a _____ from other classes. [Encapsulation]

Properties, Methods, Events, Fields, and Constructors

Many terms are important to understand when using object-oriented programming. An object is an item that can be used in code. By understanding terms such as properties, methods, events, fields, and constructors, developers can use object-oriented programming to develop applications.

Purpose

Upon completing this project, you will better understand properties, methods, events, fields, and constructors.

Steps for Completion

1. What is a property?
 - a. _____.
2. What is a method?
 - a. _____.
 - _____.
3. What is an event?
 - a. _____.
4. What is a field?
 - a. _____.
5. What is a constructor?
 - a. _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 3

Topic: Classes

Subtopic: Properties, Methods, Events, Fields, and Constructors

Objectives covered

3 Object-Oriented Programming

3.1 Construct, analyze, and use classes

3.1.1 Properties, methods, events, fields, and constructors

Creating and Using Classes

Classes are blueprints for objects. Before objects can be used in programming, developers must define classes. Understanding how to create and use classes can help developers effectively use object-oriented programming.

Purpose

Upon completing this project, you will better understand how to create and use classes.

Steps for Completion

1. What do classes contain?
 - a. _____.
2. What do classes often contain that use an object to drive action?
 - a. _____.
3. What usually happens after a class is created?
 - a. _____.
4. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ Using camel casing for properties and methods is a usual best practice.
 - b. _____ There is a limited number of objects that can be created from a class.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 3

Topic: Classes

Subtopic: Create Classes; Use Classes in Code

Objectives covered

3 Object-Oriented Programming

3.1 Construct, analyze, and use classes

3.1.2 How to create classes

3.1.3 How to use classes in code

Access Modifiers and Instantiation

Access modifiers are useful tools to be used with classes. They can protect code by controlling access to properties in a class. While access modifiers protect classes, instantiation allows one to use a class to accomplish tasks in object-oriented programming.

Purpose

Upon completing this project, you will better understand access modifiers and instantiation.

Steps for Completion

1. What is the purpose of a private property?
 - a. _____

_____.
2. What does the get accessor do?
 - a. _____
_____.
3. What does the set accessor do?
 - a. _____.
4. How are public and private properties normally capitalized?
 - a. _____

_____.
5. What is Instantiation?
 - a. _____
_____.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 3

Topic: Classes

Subtopic: Access Modifiers;
Instantiation

Objectives covered

3 Object-Oriented Programming

3.1 Construct, analyze, and use
classes

3.1.4 Access modifiers

3.1.5 Instantiation

Static Classes, Encapsulation, and Composition

While instantiating a class is useful in object-oriented programming, it may not always be necessary. Static classes can save developers time when writing code. Like access modifiers, encapsulation helps to protect classes. Composition allows for classes to be coupled and have attributes in one class be available in another class.

Purpose

Upon completing this project, you will better understand static classes, encapsulation, and composition

Steps for Completion

1. What is a static class?
 - a. _____

2. What is encapsulation?
 - a. _____

3. What is an internal class restricted to?
 - a. _____

4. How do classes share properties and attributes through composition?
 - a. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 3

Topic: Classes

Subtopic: Static vs. Instance;
Encapsulation; Composition

Objectives covered

3 Object-Oriented Programming

3.1 Construct, analyze, and use classes

3.1.6 Static vs. instance

3.1.7 Encapsulation

3.1.8 Composition

Domain 3

Lesson 2

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 3 Lesson 2, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. Classes can serve as base classes for other classes or derived classes in _____ terms. [Base Class to Derived Class]
2. A generic class is a class that does not have a particular _____ type. [Generic Classes]
3. An abstract class must have at least one abstract _____. [Abstract Classes]
4. _____ is the ability of derived classes to share functionality with base classes and have their own behaviors. [Extending Class Functionality]
5. The _____ keyword allows a method to be overridden as needed when creating classes based on a particular class. [Overriding Methods]
6. Interfaces can create empty _____ and properties that can then be used as a basis for classes. [Interfaces]
7. Overloading occurs when the same _____ is used more than once but in different ways. [Overloading]

Base vs. Derived Classes and Generic vs. Abstract Classes

Using base classes and derived classes allows developers to save time when coding. A generic class does not have a specific data type until an instance of the class is created. An abstract class is a class that has a method that is declared but not implemented. By understanding different class types, developers can efficiently code applications to perform desired actions.

Purpose

Upon completing this project, you will better understand base classes, derived classes, abstract classes, and generic classes.

Steps for Completion

1. What is a derived class?
 - a. _____
_____.
2. Why might one choose to use a generic class?
 - a. _____

_____.
3. Why might one choose to use an abstract class?
 - a. _____

_____.
4. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ A base class is built from a derived class.
 - b. _____ A generic class does not offer versatility.
 - c. _____ Angle brackets indicate a generic class.
 - d. _____ An abstract class creates flexibility among derived classes.
 - e. _____ An abstract class can be instantiated.
 - f. _____ An abstract class must have at least one abstract method.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 3

Topic: Inheritance

Subtopic: Base Class to Derived Class; Generic Classes; Abstract Classes

Objectives covered

3 Object-Oriented Programming

3.2 Construct and analyze code that uses inheritance

3.2.1 Inherit the functionality of a base class into a derived class

3.2.2 Generic classes

3.2.3 Abstract classes

Extending Class Functionality and Overriding Methods

Polymorphism allows developers to use base classes and derived classes efficiently. By understanding the concept of polymorphism, developers will know when to inherit classes from base classes and override methods.

Purpose

Upon completing this project, you will better understand polymorphism and how to use base classes and derived classes in code.

Steps for Completion

1. What is polymorphism?
 - a. _____

_____.
2. What is the main purpose of having a class inherit from a base class?
 - a. _____

_____.
3. What is the purpose of overriding a method?
 - a. _____

_____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 3

Topic: Polymorphism

Subtopic: Extending Class

Functionality; Overriding Methods

Objectives covered

3 Object-Oriented Programming

3.3 Construct and analyze code that uses polymorphism

3.3.1 Extending the functionality of a class after inheriting from a base class

3.3.2 Overriding methods in the derived class

Interfaces and Overloading

In addition to inheriting from base classes and overriding methods, polymorphism includes interfaces and overloading. Interfaces are a basis for classes. Overloading can save developers time and help with consistency in code writing.

Purpose

Upon completing this project, you will better understand interfaces and overloading.

Steps for Completion

1. What can interfaces create?
 - a. _____

 _____.
2. When does overloading occur?
 - a. _____
 _____.
3. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ Interfaces are used to define classes.
 - b. _____ For a class to use an interface, it must use all the methods declared in the interface
 - c. _____ Overloading results in more method names.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 3

Topic: Polymorphism

Subtopic: Interfaces; Overloading

Objectives covered

3 Object-Oriented Programming

3.3 Construct and analyze code that uses polymorphism

3.3.3 Interfaces

3.3.4 Overloading

Domain 4

Lesson 1

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 4 Lesson 1, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. Most applications that are developed now are _____-based. [HTML5, CSS3, and JavaScript]
2. To begin constructing and analyzing web applications, one first needs to understand the major _____ used in web applications. [HTML5, CSS3, and JavaScript]
3. In the Developer Tools, any messages one needs to see will display in the _____. [Browser Developer Tools]
4. Part of building _____ is making Hypertext Transfer Protocol (HTTP) requests and then getting responses from those requests to use information from the requests inside a web application. [HTTP Request or Response]
5. A common example of state management is ensuring that a submit button is _____ until all fields on a form are filled out. [State Management]
6. Cookies are one way that information given to or obtained from a website can be stored _____ on one's computer. [Cookies]
7. Session storage _____ when a web session ends. [Local and Session Storage]
8. ASP.NET is a leading _____ technology for web applications. [Page Lifecycle]
9. Each webpage built in ASP.NET has a distinct _____ that allows one to build controls to fire as a webpage loads and renders. [Page Lifecycle]
10. When designing an application, pay attention to _____ people need to take. [Event Model]
11. Client-side programming performs processing using languages such as _____. [Client-Side vs. Server-Side Programming]

HTML, CSS, and JavaScript

Developers need to know how web applications are managed on web servers and the types of web services that can be used. The major languages used for web applications are Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript. Each of these languages serves a different function to make web applications work.

Purpose

Upon completing this project, you will better understand HTML, CSS, and JavaScript.

Steps for Completion

1. What is the current version of JavaScript?
 - a. _____.
2. What is the purpose of HTML?
 - a. _____.
3. What is the purpose of CSS?
 - a. _____.
4. What is the purpose of JavaScript?
 - a. _____.
5. Which language is dynamic?
 - a. _____.
6. Which languages are static?
 - a. _____.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 4

Topic: Construct and Analyze Web Applications

Subtopic: HTML5, CSS3, and JavaScript

Objectives covered

4 Web Applications

4.1 Construct and analyze web applications

4.1.1 HTML5, CSS3, and JavaScript ES6

Developer Tools

Browser developer tools can be used to test and run code and troubleshoot issues that occur. In the Developer Tools, one can insert a breakpoint into code. One can also place a watch on certain components to ensure they are getting the correct values.

Purpose

Upon completing this project, you will better understand Developer Tools.

Steps for Completion

1. When using Microsoft Edge, which key on the keyboard is used to open the Developer Tools?
 - a. _____
2. How can one insert a breakpoint in code while in the Sources section of the Developer Tools?
 - a. _____
3. What does a breakpoint do?
 - a. _____
4. Which two components can one place a watch on in the Developer Tools?
 - a. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Construct and Analyze Web Applications

Subtopic: Browser Developer Tools

Objectives covered

4 Web Applications

4.1 Construct and analyze web applications

4.1.2 Browser developer tools

HTTP Requests, Responses, and State Management

HTTP requests are used to retrieve information from a server. The server will then send a response that indicates the status of the webpage. Understanding possible HTTP responses can help one know if there is a problem when a webpage connects to a server and what the specific issue might be. Another aspect of constructing web applications is state management, which is useful for web applications with forms.

Purpose

Upon completing this project, you will better understand HTTP requests, HTTP responses, and state management.

Steps for Completion

1. What is an asynchronous task?

a. _____
_____.

2. Match the HTTP request response range to its definition.

100-199	200-299	300-399	400-499	500-599
---------	---------	---------	---------	---------

- a. _____ Indicates a redirect
- b. _____ Is considered good
- c. _____ Indicates a client error
- d. _____ Indicates a server error
- e. _____ Is informational

3. What is state management?

a. _____
_____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Construct and Analyze Web Applications

Subtopic: HTTP Request or Response; State Management

Objectives covered

4 Web Applications

4.1 Construct and analyze web applications

4.1.3 HTTP request or response

4.1.4 State management

Cookies, Local Storage, and Session Storage

Cookies allow information to be stored after a web session has expired, such as when items in a website's shopping cart are still in the cart when one revisits the website later. In addition, local and session storage provide ways to store data from a website.

Purpose

Upon completing this project, you will better understand cookies, local storage, and session storage.

Steps for Completion

1. What are cookies?
 - a. _____

_____.
2. From where does a name value stored as a cookie usually come?
 - a. _____.
3. What is local storage?
 - a. _____.
4. What is session storage?
 - a. _____
_____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Construct and Analyze Web Applications

Subtopic: Cookies; Local and Session Storage

Objectives covered

4 Web Applications

4.1 Construct and analyze web applications

4.1.5 Cookies

4.1.6 Local and session storage

ASP.NET Lifecycle

ASP.NET is a technology that can be used to develop web applications. ASP.NET allows one to build controls that fire as a webpage loads and renders. ASP.NET has eight stages within its lifecycle.

Purpose

Upon completing this project, you will better understand ASP.NET.

Steps for Completion

1. Describe the first stage of the ASP.NET lifecycle.

- a. _____
_____.

2. Describe the second stage of the ASP.NET lifecycle.

- a. _____
_____.

3. Describe the third stage of the ASP.NET lifecycle.

- a. _____
_____.

4. Describe the fourth stage of the ASP.NET lifecycle.

- a. _____
_____.

5. Describe the fifth stage of the ASP.NET lifecycle.

- a. _____.

6. Describe the sixth stage of the ASP.NET lifecycle.

- a. _____
_____.

7. Describe the seventh stage of the ASP.NET lifecycle.

- a. _____
_____.

8. Describe the eighth stage of the ASP.NET lifecycle.

- a. _____
_____.

9. What is a postback?

- a. _____.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 4

Topic: Construct and Analyze Web Applications

Subtopic: Page Lifecycle

Objectives covered

4 Web Applications

4.1 Construct and analyze web applications

4.1.7 Page lifecycle

Event Models and Programming Types

An event model is a conceptual idea. It is a framework that can help developers understand what features to focus on when building code for applications. When building code, it is important to understand the difference between client-side and server-side programming to make the best choice for the application's needs.

Purpose

Upon completing this project, you will better understand what an event model is and the difference between client-side and server-side programming.

Steps for Completion

1. What is an event model?
 - a. _____

 _____.
2. Where is processing performed in client-side programming?
 - a. _____

 _____.
3. What is an advantage to using server-side programming rather than client-side programming?
 - a. _____

 _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Construct and Analyze Web Applications

Subtopic: Event Model; Client-Side vs. Server-Side Programming

Objectives covered

4 Web Applications

4.1 Construct and analyze web applications

4.1.8 Event model

4.1.9 Client-side vs. server-side programming

Domain 4

Lesson 2

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 4 Lesson 2, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. A virtual directory is a directory that maps to a physical directory on a _____. [Create Virtual Directories and Websites]
2. A real-life website has a(n) _____ page, images, and other media. [Publish Web Applications]
3. Virtual directories hide the real directories from those using the website, which is a key to _____ the website. [Web Server Role]
4. Web service messages are transmitted over _____ and use Extensible Markup Language (XML) to communicate between computers. [Client Application Consumption]
5. Web service messages can be transmitted over multiple _____. [Client Application Consumption]
6. Web _____ can be used to make casing consistent for data. [Accessing Web Services]
7. Data from a _____ request is often returned in a JavaScript Object Notation (JSON) file. [JSON]
8. One advantage of JSON data is that it can easily be _____ into other formats, such as XML. [JSON]
9. GET requests retrieve data through _____ communications. [REST API]
10. Many websites offer OAuth as a way to use familiar _____ to join other sites. [OAuth]
11. The most common way data is _____ from a web service is through Extensible Markup Language (XML). [XML]
12. Semi-structured data has a defined set of objects, but not all the objects have the same _____. [XML]
13. The design of a model-view-controller (MVC) starts with a _____, the piece of an application with which the user will interact. [MVC]
14. Model-view-viewmodel (MVVM) is best used for simple, single-page applications where there is not a lot of _____ to send back and forth. [MVVM]
15. It is not practical to hide _____ on webpages. [SPA]

Creating Websites

Developers must understand the process of publishing a website so that all aspects of the website are customized as needed. Websites can be published using web servers, and within the web servers, there are options developers can use to publish the website in the manner that is best suited for the website.

Purpose

Upon completing this project, you will better understand how to create a website.

Steps for Completion

1. What two tasks usually need to occur when one is ready to publish a website onto a web server?
 - a. _____.
2. When creating a website, what checkbox must be cleared to ensure the website is not published right away?
 - a. _____.
3. When creating a website, what should the assigned port be changed to if one is only testing the website?
 - a. _____.
4. What is the purpose of a virtual directory?
 - a. _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Configure Web Hosting

Subtopic: Create Virtual Directories and Websites

Objectives covered

4 Web Applications

4.2 Describe and configure web hosting

4.2.1 Creating virtual directories and websites

Web Servers and Publishing Web Applications

When one has made a web application, one may choose to create a test version before publishing a version that everyone can access. By understanding how to prepare the application's files and how to use a web server, developers can create test versions of a web application and alter settings for the live version of the application as needed.

Purpose

Upon completing this project, you will better understand how to publish a web application and use a web server.

Steps for Completion

1. What is the first step of publishing a test version of an application?
 - a. _____

 _____.
2. When is a website ready to be started?
 - a. _____
 _____.
3. Besides virtual directories, what is the key to controlling access to a website?
 - a. _____
 _____.
4. On a web server with File Transfer Protocol (FTP) turned on, what can one control?
 - a. _____
 _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Configure Web Hosting

Subtopic: Publish Web

Applications; Web Server Role

Objectives covered

4 Web Applications

4.2 Describe and configure web hosting

4.2.2 Publishing web applications

4.2.3 Describe the role of the web server

Web Services

Web services can be used by client applications to communicate between devices. This communication allows developers to minimize the amount of code needed to retrieve data from a server, saving developers time and reducing errors.

Purpose

Upon completing this project, you will better understand web services.

Steps for Completion

1. What is the purpose of web services?
 - a. _____

 _____.
2. What is Simple Object Access Protocol (SOAP)?
 - a. _____
 _____.
3. When creating a web service, why might one choose to create two web methods that return the uppercase and lowercase versions of input strings?
 - a. _____

 _____.
4. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ Web services use HTML to communicate between web services.
 - b. _____ When web services are stored on a web server, methods can only be used on select pages in a client application.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 4

Topic: Configure Web Hosting

Subtopic: Client Application Consumption; Accessing Web Services

Objectives covered

4 Web Applications

4.2 Describe and configure web hosting

4.3.1 Web services that are consumed by client applications

4.3.2 Accessing web services from client applications

JSON and REST APIs

Data from a web service request is often returned in a JavaScript Object Notation (JSON) file. Representational State Transfer Application Programming Interfaces (REST APIs) allow two devices to communicate. Understanding JSON and REST APIs can help developers create and use web services.

Purpose

Upon completing this project, you will better understand how JSON and REST APIs affect web services.

Steps for Completion

1. How does JSON store items?
 - a. _____
_____.
2. What is the purpose of REST APIs?
 - a. _____
_____.
3. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ JSON files offer a flexible way to store semi-structured data as there are no defined fields within objects.
 - b. _____ REST APIs allow two computers to communicate through SSH, primarily through GET and POST requests.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Configure Web Services

Subtopic: JSON; REST API

Objectives covered

4 Web Applications

4.3 Describe and configure web services

4.3.3 JSON

4.3.4 REST API

OAuth and XML

OAuth allows users to access websites with unique accounts but makes the sign-in process easier for users. Extensible Markup Language (XML) shares some similarities to HTML but serves a unique purpose when structuring a database.

Purpose

Upon completing this project, you will better understand OAuth and XML.

Steps for Completion

1. What is the purpose of OAuth?
 - a. _____
_____.
2. What is the main difference between XML and HTML?
 - a. _____

_____.
3. For what type of data is XML well-suited?
 - a. _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Configure Web Services

Subtopic: OAuth; XML

Objectives covered

4 Web Applications

4.3 Describe and configure web services

4.3.5 OAuth

4.3.6 XML

Architectural Patterns

Architectural patterns determine how applications are structured, and each one functions uniquely. Three common architectural patterns are model-view-controller (MVC), model-view-viewmodel (MVVM), and single-page application (SPA).

Purpose

Upon completing this project, you will better understand architectural patterns that can be used to develop applications.

Steps for Completion

1. Match the architectural pattern to its description.

Model-view-controller (MVC)	Model-view-viewmodel (MVVM)	Single-page application (SPA)
-----------------------------	-----------------------------	-------------------------------

- a. _____ The viewmodel stores presentation logic and acts as a go-between from the view to the model, though the view does not talk directly to the model.
 - b. _____ Everything in the application is on one page.
 - c. _____ The user interacts with the view, which sends alerts to controllers. When applicable, the controllers send data to a model. Then, the model tells the view that changes have been made, and then the view gets the changes and updates itself.
2. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ A view is known as a user interface.
 - b. _____ An MVVM model typically requires more code than an MVC model.
 - c. _____ An SPA should have more than one form to fill out.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Configure Web Services

Subtopic: MVC; MVVM; SPA

Objectives covered

4 Web Applications

4.3 Describe and configure web services

4.4.1 Model-view-controller (MVC)

4.4.2 Model-view-viewmodel (MVVM)

4.4.3 Single-page application (SPA)

Domain 5

Lesson 1

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 5 Lesson 1, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. A _____ is a single list with all the data in one group. [Characteristics and Capabilities]
2. A _____ has data broken into tables, and those tables are then related together as needed for queries. [Characteristics and Capabilities]
3. One way to identify the required fields for a database is to determine what data is needed on _____, as this will dictate what needs to be stored. [Database Design]
4. Once _____ are identified and grouped, one should establish entities and organize relationships between entities. [ERDs]
5. Normalization is a relational database concept where data is organized to prevent repeated data _____. [Normalization Concepts]
6. Normalization prevents problems when working with data as a database grows and _____ and reports are needed. [Normalization Concepts]
7. With indexes, searches are easier, but more _____ is used to hold the indexes. [Index Concepts]
8. One way to ensure the _____ data is entered into a given field is to add a constraint to a field. [Constraints]
9. A primary key should be determined when an _____ is first conceptualized. [Primary Key]
10. To add a foreign key, one should be in the _____ where the foreign key will reside. [Foreign Keys]
11. The default Enforce Foreign Key Constraint setting enforces _____. [Relationships]
12. _____ records are files in a foreign table with no matches in a primary key table. [Relationships]

Characteristics of Databases

Databases can be essential to an organization for storing information. There are many kinds of databases, including flat files and relational databases. Relational databases provide some advantages over flat files.

Purpose

Upon completing this project, you will better understand databases and the difference between a flat file and a relational database.

Steps for Completion

1. Explain what a database does.

a. _____

 _____.

2. Match the term to its definition.

Flat file	Relational database
-----------	---------------------

a. _____ Has data broken into tables, and those tables are related together as needed for queries

b. _____ A single list that has all the data in one group

- 3.

a. _____

 _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: Design and Normalize a Database

Subtopic: Characteristics and Capabilities

Objectives covered

5 Databases

5.1 Design and normalize a database

5.1.1 Characteristics and capabilities of database products

Database Design

The first step to implementing a database is to design the database. To design a database, one will need to identify the required fields in the database. These fields will then need to be grouped into entities. After database fields have been identified and grouped, an entity relationship diagram (ERD) can be used to continue the design process.

Purpose

Upon completing this project, you will better understand database design and ERDs.

Steps for Completion

1. What is an ERD?
 - a. _____

 _____.
2. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ One key in grouping fields is to make sure every group has fields that are related to each other.
 - b. _____ Most database formats accept field names with spaces.
 - c. _____ Some ERDs only show entity names, but most show the columns used in relationships.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: Design and Normalize a Database

Subtopic: Database Design; ERDs

Objectives covered

5 Databases

5.1 Design and normalize a database

5.1.2 Database design

5.1.3 Entity Relationship Diagrams (ERDs)

Normalization

As a database grows, it must be organized to prevent errors and find information efficiently. Normalization helps to organize a database by preventing problems that could arise later. The first and third forms of normalization are relevant to databases.

Purpose

Upon completing this project, you will better understand normalization.

Steps for Completion

1. Open the **514-normalization1.xlsx** file in your Domain 5 Student folder.
2. Add a column called **CustID** to the left of the Name column.
3. In the CustID column, number each entry in the table, starting with 1.
4. Add a column called **LastName** to the left of the Name field.
5. Enter the last names from the Name field in the LastName field.
6. Remove the last names from the Name field.
7. Rename the Name field **FirstName**.
8. Save the file as **514-normalization1-Completed** in your Domain 5 Student folder.
9. What is the purpose of the first form of normalization?

a. _____

10. What is the purpose of the third form of normalization?

a. _____

Project Details

Project file

514-normalization1.xlsx

Estimated completion time

10 minutes

Video reference

Domain 5

Topic: Design and Normalize a Database

Subtopic: Normalization Concepts

Objectives covered

5 Databases

5.1 Design and normalize a database

5.1.4 Normalization concepts (to 3NF)

Indexes and Constraints

Part of database design includes the ability to search for information. Creating an index in a database helps make information within the database easier to find. In addition, constraints can make information easier to find and easier to use.

Purpose

Upon completing this project, you will better understand indexes and constraints.

Steps for Completion

1. What is a clustered index?
 - a. _____
_____.
2. What is a non-clustered index?
 - a. _____

_____.
3. What is the purpose of a constraint?
 - a. _____.
4. Describe a situation in which one might use a constraint.
 - a. _____

_____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: Design and Normalize a Database

Subtopic: Index Concepts; Constraints

Objectives covered

5 Databases

5.1 Design and normalize a database

5.1.5 Indexes

5.1.6 Constraints

Primary and Foreign Keys

Primary and foreign keys can be used to design entities, and they each serve different purposes. Primary keys help ensure data is in first normal form. Foreign keys are useful for a database with multiple tables. Using primary and foreign keys can help ensure a database is easy to use and information is easy to find.

Purpose

Upon completing this project, you will better understand primary and foreign keys.

Steps for Completion

1. What is the purpose of a primary key field?
 - a. _____
2. What is the purpose of a foreign key field?
 - a. _____
3. What is a one-to-one relationship?
 - a. _____
4. What is a one-to-many relationship?
 - a. _____
5. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ A primary key should be determined after a database has been created.
 - b. _____ A column containing different customer ID numbers could be used as a primary key.
 - c. _____ Foreign keys are used to establish primary keys.
 - d. _____ To add a foreign key, one should be in the table where the foreign key will reside.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 5

Topic: Design and Normalize a Database

Subtopic: Primary Key; Foreign Keys

Objectives covered

5 Databases

5.1 Design and normalize a database

5.1.7 Primary key

5.1.8 Foreign keys

Relationships and Cardinality

Understanding how relationships between tables function is essential to designing a database that works efficiently. Foreign keys allow one to create relationships between tables that enforce referential integrity. Cardinality is the likelihood that values in every record will be unique field to field.

Purpose

Upon completing this project, you will better understand the relationships between tables.

Steps for Completion

1. What does the default Enforce Foreign Key Constraint setting do?
 - a. _____
2. What is referential integrity?
 - a. _____
3. What are orphaned records?
 - a. _____
4. How can orphaned records be prevented?
 - a. _____
5. In any given table, which field is most likely to have the highest cardinality among the fields within the table?
 - a. _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: Design and Normalize a Database

Subtopic: Relationships; Cardinality

Objectives covered

5 Databases

5.1 Design and normalize a database

5.1.9 Relationships

5.1.10 Cardinality

Domain 5

Lesson 2

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 5 Lesson 2, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. One hallmark of writing any code in an application is to _____ the code as much as possible. [Stored Procedures]
2. When using a SQL UPDATE statement, one always updates a _____, not a field. [Updating and Selecting Data]
3. It is very important to have a _____ clause in an UPDATE statement because without it, the UPDATE statement would update every record in the table. [Updating and Selecting Data]
4. Data Manipulation Language (DML) covers SQL keywords such as SELECT, UPDATE, and _____. [DML vs. DDL]
5. Data Definition Language (DDL) covers SQL keywords such as _____, ALTER, and DROP. [DML vs. DDL]
6. Functions use an input _____ to generate an output. [Functions]
7. The DROP keyword requires the _____ of object one is dropping and the name of the object one is dropping. [Triggers]
8. Most searches for data in SQL Server involve using a _____ clause in specifying criteria for what someone desires for a result. [Cursors]
9. Most queries run in a database will use two or more _____. [Joins]
10. There are three types of joins: left, right, and _____ outer joins. [Joins]
11. Creating a nonclustered index takes little _____. [Indexes]

Stored Procedures

Protecting code is important in a database, and stored procedures can keep code safe. Stored procedures accept both input and output parameters. Stored applications can retrieve data from a database within an application.

Purpose

Upon completing this project, you will better understand what stored procedures are and how to use them.

Steps for Completion

1. What is a stored procedure?
 - a. _____
_____.
2. What is the purpose of a stored procedure?
 - a. _____
_____.
3. What are parameters?
 - a. _____
_____.
4. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ Input parameters must be specified as input parameters.
 - b. _____ Input parameters are used in apps to return a value that the application will use later.
 - c. _____ An output parameter determines the order in which outputs are sent to an application.
 - d. _____ A stored procedure is created with a NEW PROCEDURE statement.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: SQL Queries

Subtopic: Stored Procedures

Objectives covered

5 Databases

5.2 Construct, analyze, and optimize ANSI SQL queries

5.2.1 Creating and accessing stored procedures

Database Language

It is important to understand how to make changes to the data in a database and how to make structural changes to the database to use its full capabilities. SQL keywords perform different actions on a database or data within a database. Depending on their function, SQL keywords are considered either Data Definition Language (DDL) or Data Manipulation Language (DML).

Purpose

Upon completing this project, you will better understand DDL, DML, and how to make changes to a database.

Steps for Completion

1. Match the SQL keyword to its definition.

SELECT	FROM	WHERE
ALTER	DROP	INSERT

- _____ Changes the structure of an object
 - _____ Adds records to a table
 - _____ Determines the table from which data will be used
 - _____ Acts as a filter on data
 - _____ Chooses the data
 - _____ Removes objects from a database
2. Which language is used to make changes to the information in a database?
 - _____
 3. Which language deals with the structure of a database?
 - _____

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: SQL Queries

Subtopic: Updating and Selecting Data; DML vs. DDL

Objectives covered

5 Databases

5.2 Construct, analyze, and optimize ANSI SQL queries

5.2.2 Updating and selecting data

5.2.3 DML vs. DDL

Functions and Triggers

Functions and triggers can be used in a database to perform different tasks. Functions generate an output using an input parameter. Triggers help control processes within a database and help keep data intact so that data is not inadvertently changed.

Purpose

Upon completing this project, you will better understand functions and triggers.

Steps for Completion

1. What is the purpose of functions?
 - a. _____
_____.
2. What two tasks can triggers perform?
 - a. _____

_____.
3. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ Functions run through SELECT statements.
 - b. _____ When executing a function, dbo. is needed in front of the function to tell SQL Server that one is running a database observation.
 - c. _____ Calculated columns are often stored in database tables.
 - d. _____ Triggers can convey messages to people performing actions in a database.
 - e. _____ Only administrators should have permission to drop objects from a database.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 5

Topic: SQL Queries

Subtopic: Functions; Triggers

Objectives covered

5 Databases

5.2 Construct, analyze, and optimize ANSI SQL queries

5.2.4 Functions

5.2.5 Triggers

Cursors and Indexes

Cursors can be used to scan a table line-by-line to match the criteria for the desired result. Indexes can be used to make searching a database faster. An index can be based on a primary key field or a non-primary key field.

Purpose

Upon completing this project, you will better understand cursors and indexes.

Steps for Completion

1. What is the purpose of a cursor?
 - a. _____
_____.
2. What command can be used to search one record at a time?
 - a. _____
_____.
3. When using a cursor, what happens when the fetching is complete?
 - a. _____
_____.
4. Which type of index is based on non-primary key fields?
 - a. _____
_____.
5. Which type of index is based on primary key fields?
 - a. _____
_____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: SQL Queries

Subtopic: Cursors; Indexes

Objectives covered

5 Databases

5.2 Construct, analyze, and optimize ANSI SQL queries

5.2.6 Cursors

5.2.8 Indexes

Joins

Joins can make searching for data in a database easier. Joins allow one to find data in multiple tables. One can use an inner join, left outer join, right outer join, or a full outer join to locate relevant data in a database. The type of join used will depend on what data one is looking to retrieve and where that data exists in the tables being joined.

Purpose

Upon completing this project, you will better understand joins.

Steps for Completion

1. What is the purpose of a relational database?

a. _____

 _____.

2. Match the join type to its description.

Left outer join	Right outer join	Inner join	Full outer join
-----------------	------------------	------------	-----------------

- a. _____ Returns all records from the right table and matching records from the left table
- b. _____ Returns records that have matching values in both tables
- c. _____ Returns all records when there is a match in either the left or right table
- d. _____ Returns all records from the left table and matching records from the right table

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: SQL Queries

Subtopic: Joins

Objectives covered

5 Databases

5.2 Construct, analyze, and optimize

ANSI SQL queries

5.2.7 Joins

Domain 5

Lesson 3

Software
Development

Fill-in-the-Blanks

Instructions: While watching Domain 5 Lesson 3, fill in the missing words according to the information presented by the instructor. [References are found in the brackets.]

1. A transaction in SQL can be compared to a _____ transaction. When making a deposit, no part of the transaction will go through if something goes wrong. [Commit]
2. _____ can be rolled back as needed. [Rollback]
3. Some transactions lend themselves to a _____ rollback. [Save]
4. When implementing a transaction, two people can affect the same _____ simultaneously which often hurts data. [Concurrency]
5. One way to avoid _____ is to set an isolation level for a transaction. [Isolation Levels]
6. A table with no locks has _____ that multiple people view simultaneously. [Lock]
7. An entity framework is a database _____. [Entity Framework]
8. Typically, when an application connects to a _____, the connection is opened and closed when no longer needed. [Connection Pools]
9. Language Integrated Query (LINQ) allows for more effective coding than writing _____ code inside of C#. [LINQ]
10. Document databases are known as _____ databases. [Document Databases]
11. Some argue that a true key-value database has _____ value per key, but that is more the exception rather than the rule. [Key-Value Databases]

Commit and Rollback

Transactions allow a block of code to run. Transactions can be allowed, or they can be prevented from saving to a database. One can specify criteria for a transaction to meet. If the criteria are met, the transaction will be allowed. If the criteria are not met, the transaction will be prevented from occurring.

Purpose

Upon completing this project, you will better understand how to commit and roll back a transaction.

Steps for Completion

1. What is a transaction?
 - a. _____

 _____.
2. Why might a transaction be rolled back?
 - a. _____

 _____.
3. What keyword is used to allow a transaction?
 - a. _____
 _____.
4. What keyword is used to prevent a transaction?
 - a. _____
 _____.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: Manage Transactions

Subtopic: Commit; Rollback

Objectives covered

5 Databases

5.3 Manage transactions

5.3.1 Commit

5.3.2 Rollback

Save and Concurrency

Transactions can be committed or rolled back, but saving a transaction allows one to perform a partial rollback. While saving a transaction can prevent errors with data, concurrency can cause problems within a database.

Purpose

Upon completing this project, you will better understand concurrency and the purpose of saving a transaction.

Steps for Completion

1. What is the purpose of a SAVE statement when implementing a transaction?
 - a. _____

 _____.
2. What would happen if two people ran the same transaction simultaneously?
 - a. _____
 _____.
3. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ When using a SAVE statement with a transaction, everything before the save point can be rolled back.
 - b. _____ Two people cannot run a transaction at the same time.
 - c. _____ Many SQL statements function better by themselves in a window.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: Manage Transactions

Subtopic: Save; Concurrency

Objectives covered

5 Databases

5.3 Manage transactions

5.3.3 Save

5.3.4 Concurrency

Isolation Levels and Locks

Setting an isolation level on a transaction can help prevent concurrency. Isolation levels that can be applied to a transaction include read uncommitted, read committed, repeatable read, and serializable. On the other hand, it may be better for some databases to have no lock on data.

Purpose

Upon completing this project, you will better understand different isolation levels that can be applied to transactions and why one may not want a lock on data in a database.

Steps for Completion

1. What happens when an isolation level is set on a transaction?

a. _____
_____.

2. Match the isolation level to its description.

Repeatable read	Read committed
Serializable	Read uncommitted

- a. _____ Rows being updated or deleted cannot be changed until current changes are done.
- b. _____ People can see rows that are being changed, but the transaction has yet to be committed.
- c. _____ Rows being inserted, updated, or deleted cannot be changed until current changes are done.
- d. _____ Data is locked until committed changes are made.

3. For what type of databases might one choose to ensure are not locked?

a. _____
_____.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 5

Topic: Manage Transactions

Subtopic: Isolation Levels; Lock

Objectives covered

5 Databases

5.3 Manage transactions

5.3.5 Isolation levels

5.3.6 Lock

Entity Frameworks, Connection Pools, and LINQ

Entity frameworks, connections pools, and Language Integrated Query (LINQ) are methods to access a database. Entity frameworks can be built using a database-first or code-first approach. Connection pools can provide convenient ways to connect an application to a database. LINQ can make coding more efficient.

Purpose

Upon completing this project, you will better understand entity frameworks and connection pools.

Steps for Completion

1. What is an entity framework?
 - a. _____.
2. What is the database-first approach to building an entity framework?
 - a. _____.
3. What is the code-first approach to building an entity framework?
 - a. _____.
4. What is the end goal of an entity framework?
 - a. _____.
5. What is the purpose of a connection pool?
 - a. _____.
6. Why might one want to use a connection pool?
 - a. _____.
7. What is the purpose of LINQ?
 - a. _____.
8. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ An entity framework is not time-consuming to build.
 - b. _____ The best way to test a connection pool is to test how much memory it uses.
 - c. _____ When using LINQ, statements must be in the order in which events occur.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 5

Topic: Database Access Methods

Subtopic: Entity Framework;
Connection Pools; LINQ

Objectives covered

5 Databases

5.4 Describe database access methods

5.4.1 Entity Framework (code-first and database-first)

5.4.2 Connection pools

5.4.3 LINQ

Document and Key-Value Databases

Document databases are filled with documents and have no defined keys or columns. Key-value databases have key items, also known as identifiers, and values that describe the identifiers. Understanding these different types of databases can help one know when to use which type of database.

Purpose

Upon completing this project, you will better understand document databases and key-value databases.

Steps for Completion

1. What types of databases are known as NoSQL databases?
 - a. _____

2. Determine whether each statement is true or false. Use T for true and F for false.
 - a. _____ A collection of data can be considered a database.
 - b. _____ Every database is a SQL database.
 - c. _____ Document databases are unstructured databases.
 - d. _____ XML documents are key-value databases.
 - e. _____ Key-value databases are structured databases.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 5

Topic: NoSQL Databases

Subtopic: Document Databases;
Key-Value Databases

Objectives covered

5 Databases

5.5 Describe types of NoSQL databases

5.5.1 Document databases

5.5.2 Key-value databases

Appendix

Software
Development

Glossary

Term	Definition
1NF	First Normal Form (1NF) is a form of normalization that ensures that each column holds a single piece of data within an entity.
3NF	Third Normal Form (3NF) is a form of normalization that ensures a change to data in one field in an entity does not necessitate a change in another field within a database.
Abstract Class	A base class that can have its methods overwritten when a class is derived from the base class.
Access Modifier	A keyword, such as public or private, that designates access to a class and parts of a class.
Agile	A project management method geared toward projects with rapidly changing requirements.
Array	A fixed amount of values for a single variable.
Asynchronous Tasks	Tasks that can run concurrently.
Base Class	A class that serves as the basis for other, or, derived classes.
Binary Search	A recursive search through identifying a middle element until that element matches what is being searched.
Boolean	A data type that stores a true or false value.
Branch	In the context of version control, a copy of files and folders a group can work on and then merge with other branches.
Browser Developer Tools	A suite of tools that help developers follow logic of and troubleshoot webpages from within a browser.
Bubble Sort	A sort in which each element is compared to the next element and then arranged according to the rules of the sort.
Check In	In the context of version control, the process of saving changes to a file or folder and unlocking the file or folder for editing purposes.
Check Out	In the context of version control, the process of locking a file or folder so that only one person can work the file or folder.
Class	A blueprint for an object.
Client-Side Programming	Programming involving code that is processed on a client machine when an app runs.
Clone	In the context of version control, the saving of a local copy of all or part of a repository from GitHub.
Commit	In terms of transactions, the finalizing of all processes within a transaction.
Concurrency	The act of action being taken on the same data from two different people at the same time.
Connection Pool	A group of connections an app uses to connect from the app to a server, often a database server.
Constraint	In terms of databases, a limitation on the data that can be entered within a data field.
Constructor	A block of code that forms a class.
Cookie	A small file that stores, on a local computer, information from a web session that can be reused in a later session.
CSRF	Cross-Site Request Forgery (CSRF) is a malicious mechanism in which an attacker uses a person's authentication information to force the person to do malicious activity on a website.
CSS	Cascading Style Sheets (CSS) is a formatting language that controls how information is formatted on a webpage.
Cursor	A SQL keyword used to run through a table, one line at a time.
Database	A file or series of files that store information and allow people to develop queries and reports based off the information.
DDL	Data Definition Language (DDL) consists of SQL keywords used to control the structure of a database.
Decision Table	A matrix that shows actions to take based on one or more conditions.
Decryption	The undoing of encryption to data, done through using a decryption key.

Term	Definition
Defensive Coding	A method of coding to where code that is not needed is given consideration for removal and coding is done in such a way as to handle errors gracefully.
Delegate	A block of code that protects a function that should not be exposed to users within an application.
Deployment	The phase in the SDLC in which an app is distributed, often through an installation package.
Derived Class	A class that is based off another class.
Dictionary	A data source that stores data in key-value pairs.
Digital Signature	A hashing of data to track whether the data is altered as it goes from a source to a destination.
DML	Data Manipulation Language (DML) consists of SQL keywords used to retrieve or change data within a database.
Do While Loop	A block of code that runs while a condition is true but runs at least once regardless of whether the condition specified for the loop is true.
Document Database	A database that has records that consist of entire documents.
Double	A data type that stores decimal numbers.
Encapsulation	The act of shielding one item from another item.
Encryption	The scrambling of data to protect it from unwanted views.
Entity Framework	A framework, built either from code or from a database, that an app uses as a database map.
ERD	An Entity Relationship Diagram (ERD) is a diagram that shows, for a database, entities and relationships between those entities.
Event	An action such as a mouse click, a window opening, or a text field changing.
Event	A happening that often causes a method to run, such as a mouse click or a window opening or a text field changing.
Event Model	A framework in which control in an app is driven by events.
Expression	In the context of math, a calculation with multiple operands.
Flat File	A type of database to where all the data is stored in a single file.
Flowchart	A shape-filled chart, used to represent the logic of a program.
For Loop	A block of code that runs a set number of times.
Foreign Key	A field in a table used to create a relationship to another table.
Function	In the context of databases, a block of code used to return the result of a calculation on data.
Garbage Collection	A form of automatic memory management that frees up memory space when it is no longer needed.
Generic Class	A class that can use multiple data types.
GitHub	A repository for version control on code.
Hashing	A form of encrypting data to where the data cannot be decrypted.
HTML5	Hypertext markup language (HTML) is a markup language that controls how information displays on a webpage.
HTTP Request	Code that requests information from and the status of a web app on a server.
HTTP Response	A numbered response from a web server indicating that status of an HTTP request.
If Else Statement	A decision structure that checks for a condition and runs a block of code if the condition is true and a different block of code if the condition is false.
If Statement	A decision structure that checks for a condition and runs a block of code if the condition is true.
IFrame	A frame inside an HTML file that displays the contents of another HTML file.
Immutability	The act of having a variable that has a value that cannot be changed once it is set.
Implementation	The phase in the SDLC in which the actual code writing takes place.
Index	In terms of databases, a method of storing data to make data easier to find when conducting searches on the data.

Term	Definition
Inner Join	A type of join in which only records that are common to both sides of a join appear in a result set from a query.
Instance	In the context of a class, an object that has been created from the class.
Instantiation	The creation of an object based off a class.
Integer	A data type that stores whole numbers.
Interface	In the context of classes, a starting point with methods from which many classes can be built using those methods.
Isolation Level	The level in which data is protected from being seen or changed while a transaction on the data is taking place.
JavaScript	A scripting language that adds dynamic elements to a webpage.
Join	A process by which two tables are brought together to display related data within a query.
JSON	JavaScript Object Notation (JSON) is a data storage format that stores information in key-value pairs.
Key-Value Database	A database that stores data in key-value pairs.
Linear Search	A sequential search through a list until a list item is found.
Linked Lists	An item list that allows for changing the number of values in a list as needed and can support multiple data types within the list.
LINQ	Language-Integrated Query (LINQ) allows a developer to integrate query language into C#.
Local Storage	In the context of web browsing, information that is taken from a webpage interaction and stored on a local computer.
Lock	The ability to protect data from being viewed or changed while it is being changed.
Maintenance	The phase in the SDLC where bugs in an app are identified and fixed and enhancement requests are accepted for an app.
Memory Heap	An adjustable amount of memory set aside to store items such as objects.
Memory Stack	A block of memory that uses a fixed memory allocation to store data.
Merge	In the context of version control, the combining of changes made by multiple people to a single file.
Merge Sort	A sort in which data is broken up into small sets, sorted within the sets, and then merged.
Method	A function that takes action on an object.
MVC	Model-View-Controller (MVC) is a design model in which a model stores data, a view updates data people see, and a controller manipulates data.
MVVM	Model-View-ViewModel (MVVM) is a design model in which a model stores business logic and data, the view and viewmodel contain presentation and presentation logic, and data binding occurs between the view and viewmodel.
Normalization	The process of taking specific steps to ensure data is consistent throughout a database.
OAuth	Open Authorization (OAuth) is an authentication mechanism that allows a person to use one username/password combination to access multiple sites.
Object	A programmable item in an app.
Outer Join	A type of join in which all records from one table and related records from a related table appear in a result set from a query.
Overloading	The act of coding two or more methods with the same name but with different functionality.
Overriding	In the context of classes, changing a method in a derived class to make it differ from the base class from which it originates.
Page Lifecycle	A series of steps for running a webpage within a web app.
Planning and Design	The phase in the SDLC in which the technical and user interface designs take place.
Polymorphism	The ability of derived classes to share functionality with base classes and also have their own behaviors.

Term	Definition
Primary Key	A field (or fields) in a table in which the values entered in the field or fields is unique, thus ensuring every record in a table is unique.
Private Key	A key used usually to decrypt data.
Promise	In the context of coding, the act of making sure every function in a code block has a purpose.
Property	A characteristic on an object.
Pseudocode	An English-language outline for eventual code examples in an app.
Public Key	An encryption key used by an app to send to a user to encrypt data.
Queue	An item list that operates on a first-in-first-out basis.
Quick Sort	A sort in which an element becomes a pivot point and then elements are placed to the left or right of the pivot element accordingly.
Recursion	A command within a function that calls itself to run itself if a condition is met.
Relational Database	A type of database to where data is stored in separate entities and then columns from those entities are related to each other as needed.
Relationship	A connection between two tables in a database that allows one table to look up information from another table based on a common field between the two tables.
Requirement Analysis	The phase in the SDLC that is the process of detailing the business requirements for an app.
REST API	A Representational State Transfer Application Programming Interface (Restful API) allows two computers to communicate through HTTP.
Rollback	In the context of version control, the discarding of uncommitted changes to a file. In the context of transactions, the undoing of a transaction before it is committed.
Save	In terms of transactions, the ability to roll back a transaction to a designated point within the transaction.
SDLC	The Software Development Life Cycle (SDLC) is a software development method consisting of several phases used in the development process.
SELECT	A statement used to retrieve data from a database.
Selection Sort	A sort in which each element is sorted one by one and slotted accordingly within a list.
Server-Side Programming	Programming involving code that is processed on a server when an app runs.
Session Storage	In the context of web browsing, information that is stored and can be reused from a single web session.
Shared Key	An encryption key used by two parties. One party uses the key to encrypt data and the other to decrypt data.
SOAP	Simple Object Access Protocol (SOAP) is a protocol used to exchange structured information between two computers.
SPA	A single page application (SPA) is a web app in which all the information and presentation is done within a single webpage.
SQL Injection	The adding of SQL code to a data form with the intent that the code will steal or cause harm to data in a SQL database.
Stack	In the context of reading errors, the order in which errors occur, along with the line numbers containing the code causing the error. In the context of lists, a last-in-first-out method of managing items on lists.
State Management	The act of dealing with the state of objects on a form (enabled or disabled, for example) as a person interacts with a form on a webpage.
Static Method	A portion of a class that is held as a value every time an object is created from the class.
Stored Procedure	A saved query used to retrieve or change information within a database.
String	A data type that stores text.

Term	Definition
Switch	A decision structure used to check for multiple conditions and then run the block of code that matches the condition being true.
Synchronous Tasks	Tasks that happen one after the other.
Testing	The phase in the SDLC in which security, permissions, and functionality for an app are verified and fixed where necessary.
Throwing Exception	A line of code that causes an error to occur and allows a developer to show the properties of the error.
Trigger	A mechanism for inciting action on a database, either in the form of driving action after an update has been made or to prevent a change from being made.
Try-Catch-Finally	An exception handling block of code to where code tries to run and if it cannot run without causing an error, a catch block runs. The finally block runs after either the try or catch block completes.
Unit Testing	The testing of a block of code to ensure it functions properly.
UPDATE	A SQL keyword used to change data in a database.
Virtual Directory	In the context of websites, a pointer to an actual directory that holds files and folders for the website.
While Loop	A block of code that runs while a condition is true.
XML	eXtensible Markup Language (XML) is a markup language used to organize data within a document.

Objectives

Software Development Objectives				
Domain 1 Core Programming Concepts	Domain 2 Software Development Principles	Domain 3 Object-Oriented Programming	Domain 4 Web Applications	Domain 5 Databases
1.1 Describe computer storage and data types 1.1.1 How a computer stores programs and instructions in computer memory 1.1.2 Memory stacks and heaps 1.1.3 Memory size requirements for various data storage types, numeric data, and textual data 1.1.4 Garbage collection	2.1 Describe Software Development Life Cycle (SDLC) management 2.1.1 Requirement analysis 2.1.2 Planning and design 2.1.3 Implementation 2.1.4 Testing 2.1.5 Deployment 2.1.6 Maintenance 2.1.7 Agile concepts	3.1 Construct, analyze, and use classes 3.1.1 Properties, methods, events, fields, and constructors 3.1.2 How to create classes 3.1.3 How to use classes in code 3.1.4 Access modifiers 3.1.5 Instantiation 3.1.6 Static vs. instance 3.1.7 Encapsulation	4.1 Construct and analyze web applications 4.1.1 HTML5, CSS3, and JavaScript ES6 4.1.2 Browser developer tools 4.1.3 HTTP request or response 4.1.4 State management 4.1.5 Cookies 4.1.6 Local and session storage 4.1.7 Page lifecycle 4.1.8 Event model 4.1.9 Client-side vs. server-side programming	5.5 Design and normalize a database 5.1.1 Characteristics and capabilities of database products 5.1.2 Database design 5.1.3 Entity Relationship Diagrams (ERDs) 5.1.4 Normalization concepts (to 3NF) 5.1.5 Indexes 5.1.6 Constraints 5.1.7 Primary key 5.1.8 Foreign keys 5.1.9 Relationships
1.2 Construct and analyze algorithms and flowcharts to solve programming problems 1.2.1 Decision structures used in all computer programming languages 1.2.2 If decision structures 1.2.3 Multiple decision structures, such as if...else and switch 1.2.4 Reading and constructing flowcharts 1.2.5 Decision tables 1.2.6 Evaluating expressions 1.2.7 For loops 1.2.8 While loops 1.2.9 Do...while loops 1.2.10 Recursion	2.2 Interpret application specifications 2.2.1 Reading application specifications and translating them into prototypes and code 2.2.2 Selecting appropriate application type and components	3.2 Construct and analyze code that uses inheritance 3.2.1 Inherit the functionality of a base class into a derived class 3.2.2 Generic classes 3.2.3 Abstract classes	4.2 Describe and configure web hosting 4.2.1 Creating virtual directories and websites 4.2.2 Publishing web applications 4.2.3 Describe the role of the web server	5.2 Construct, analyze, and optimize ANSI SQL queries 5.2.1 Creating and accessing stored procedures 5.2.2 Updating and selecting data 5.2.3 DML vs. DDL 5.2.4 Functions 5.2.5 Triggers 5.2.6 Cursors 5.2.7 Joins 5.2.8 Indexes
1.3 Incorporate error handling into applications or modules 1.3.1 Structured exception handling (try-catch-finally) 1.3.2 Unit testing 1.3.3 Throwing exceptions 1.3.4 Reading the stack 1.3.5 Defensive coding 1.3.6 Understand scoping in exception handling	2.3 Construct and analyze code that uses algorithms and data structures 2.3.1 Arrays 2.3.2 Stacks 2.3.3 Queues 2.3.4 Linked lists 2.3.5 Dictionaries (key-value pairs)	3.3 Construct and analyze code that uses polymorphism 3.3.1 Extending the functionality of a class after inheriting from a base class 3.3.2 Overriding methods in the derived class 3.3.3 Interfaces 3.3.4 Overloading	4.3 Describe and configure web services 4.3.1 Web services that are consumed by client applications 4.3.2 Accessing web services from client applications 4.3.3 JSON 4.3.4 REST API	5.3 Manage transactions 5.3.1 Commit 5.3.2 Rollback 5.3.3 Save 5.3.4 Concurrency 5.3.5 Isolation levels 5.3.6 Lock

Software Development Objectives

Domain 1 Core Programming Concepts	Domain 2 Software Development Principles	Domain 3 Object-Oriented Programming	Domain 4 Web Applications	Domain 5 Databases
	2.3.6 Sorting algorithms (selection sort, bubble sort, quick sort, and merge sort) 2.3.7 Searching algorithms (linear search and binary search) 2.3.8 Performance implications of various data structures 2.3.9 Choosing the right data structure		4.3.5 OAuth 4.3.6 XML	
1.4 Construct and analyze code based on functional programming patterns 1.4.1 Event 1.4.2 Delegate 1.4.3 Promises 1.4.4 Synchronous vs. asynchronous programming (AJAX and XHR) 1.4.5 Immutability	2.4 Describe the purpose of version control systems 2.4.1 GitHub 2.4.2 Check-in 2.4.3 Check-out 2.4.4 Merge 2.4.5 Branch 2.4.6 Rollback 2.4.7 Clone 2.4.8 Resolving conflicts		4.4 Describe and identify architectural patterns' 4.4.1 Model-view- controller (MVC) 4.4.2 Model-view- viewmodel (MVVM) 4.4.3 Single page application (SPA)	5.4 Describe database access methods 5.4.1 Entity Framework (code-first and database-first) 5.4.2 Connection pools 5.4.3 LINQ
	2.5 Describe secure coding concepts 2.5.1 Encryption, hashing and digital signatures 2.5.2 Public, private, and shared keys 2.5.3 Mitigating cross-site request forgery (CSRF) 2.5.4 SQL injection 2.5.5 Risks of using IFrames			5.5 Describe types of NoSQL databases 5.5.1 Document databases 5.5.2 Key-value databases

Lesson Plan

Approximately 25.5 hours of video, labs, and projects

Software
Development

Domain 1 Lesson Plan

Domain 1 - Core Programming Concepts [approximately 5.5 hours of videos, labs, and projects]				
Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
Pre-Assessment Assessment time - 00:30:00	Core Programming Concepts: Pre-Assessment			
Lesson 1 Video time - 00:08:27 Exercise Lab time - 00:12:00 Workbook time - 00:25:00	Computer Storage and Data Types How to Study for This Exam How Things Are Stored Memory Stacks and Heaps Memory Size Requirements Garbage Collection	1.1 Describe computer storage and data types 1.1.1 How a computer stores programs and instructions in computer memory 1.1.2 Memory stacks and heaps 1.1.3 Memory size requirements for various data storage types, numeric data, and textual data 1.1.4 Garbage collection	Compiling Code Stacks and Heaps Garbage Collection Code	Reading and Storing Code – pg. 9 N/A Memory Stacks and Heaps – pg. 10 N/A Memory Size Requirements – pg. 11 N/A Garbage Collection – pg. 12 Program.cs
Lesson 2 Video time - 00:14:55 Exercise Lab time - 00:16:00 Workbook time - 01:00:00	Algorithms and Flowcharts Decision Structures If Statements Multiple Decision Structures Flowcharts Decision Tables Evaluating Expressions For Loops While Loops Do...While Loops Recursion	1.2 Construct and analyze algorithms and flowcharts to solve programming problems 1.2.1 Decision structures used in all computer programming languages 1.2.2 If decision structures 1.2.3 Multiple decision structures, such as if...else and switch 1.2.4 Reading and constructing flowcharts 1.2.5 Decision tables 1.2.6 Evaluating expressions 1.2.7 For loops 1.2.8 While loops 1.2.9 Do...while loops 1.2.10 Recursion	Understanding If Statements If...Else Statements Else If Statements For Loops	Decision Structures – pg. 15 N/A Multiple Decision Structures – pg. 16-17 N/A Flowcharts and Decision Tables – pg. 18 N/A Evaluating Expressions – pg. 19 N/A For Loops – pg. 20 N/A While and Do...While Loops – pg. 21 N/A Recursion – pg. 22 N/A
Lesson 3 Video time - 00:13:40 Exercise Lab time - 00:16:00 Workbook time - 00:55:00	Error Handling Structured Exception Handling Unit Testing Throwing Exceptions Reading Stacks Defensive Coding Scoping in Exception Handling Functional Programming Patterns Event Delegate Promises Synchronous vs.	1.3 Incorporate error handling into applications or modules 1.3.1 Structured exception handling (try-catch-finally) 1.3.2 Unit testing 1.3.3 Throwing exceptions 1.3.4 Reading the stack 1.3.5 Defensive coding 1.3.6 Understand scoping in exception handling 1.4 Construct and analyze code based on functional programming patterns 1.4.1 Event 1.4.2 Delegate	Understanding Structured Exception Handling Managing Unused Code Understanding When Code Runs	Structured Exception Handling – pg. 25 N/A Unit Testing and Exceptions – pg. 26 N/A Reading Stacks and Defensive Coding – pg. 27 N/A Scoping in Exception Handling – pg. 28 N/A Events and Delegates – pg. 29

Domain 1 - Core Programming Concepts [approximately 5.5 hours of videos, labs, and projects]

Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
	Asynchronous Immutability	1.4.3 Promises 1.4.4 Synchronous vs. asynchronous programming (AJAX and XHR) 1.4.5 Immutability		141-event.html Promises – pg. 30 143-promises.html Synchronous vs. Asynchronous – pg. 31 144-synchronous vs. asynchronous.txt Immutability – pg. 32 N/A
Post-Assessment Assessment time - 01:00:00	Core Programming Concepts: Post- Assessment			

Domain 2 Lesson Plan

Domain 2 - Software Development Principles [approximately 6 hours of videos, labs, and projects]				
Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
Pre-Assessment Assessment time - 00:30:00	Software Development Principles: Pre-Assessment			
Lesson 1 Video time - 00:09:23 Exercise Lab time - 00:00:00 Workbook time - 00:45:00	SDLC Management Requirement Analysis Planning and Design Implementation Testing Deployment Maintenance Agile Concepts Interpret Application Specifications Read Application Specifications Select Appropriate Types and Components	2.1 Describe Software Development Life Cycle (SDLC) management 2.1.1 Requirement analysis 2.1.2 Planning and design 2.1.3 Implementation 2.1.4 Testing 2.1.5 Deployment 2.1.6 Maintenance 2.1.7 Agile concepts 2.2 Interpret application specifications 2.2.1 Reading application specifications and translating them into prototypes and code 2.2.2 Selecting appropriate application type and components	N/A	Requirement Analysis – pg. 35 211-requirements.docx Planning, Design, and Implementation – pg. 36 N/A Testing and Deployment – pg. 37 Program.cs Maintenance – pg. 38 216-Maintenance.docx Agile Concepts – pg. 39 N/A Read Application Specifications – pg. 40 N/A Select Appropriate App Types – pg. 41 N/A
Lesson 2 Video time - 00:12:06 Exercise Lab time - 00:12:00 Workbook time - 01:05:00	Code With Algorithms and Data Structures Arrays Stacks Queues Linked Lists Dictionaries Sorting Algorithms Searching Algorithms Performance Implications Choosing the Best Data Structure	2.3 Construct and analyze code that uses algorithms and data structures 2.3.1 Arrays 2.3.2 Stacks 2.3.3 Queues 2.3.4 Linked lists 2.3.5 Dictionaries (key-value pairs) 2.3.6 Sorting algorithms (selection sort, bubble sort, quick sort, and merge sort) 2.3.7 Searching algorithms (linear search and binary search) 2.3.8 Performance implications of various data structures 2.3.9 Choosing the right data structure	Understanding Arrays Understanding Queues Understanding Searches	Arrays – pg. 44 N/A Stacks – pg. 45 Program.cs Queues – pg. 46 Program.cs Linked Lists and Dictionaries – pg. 47 Program.cs Sorting Algorithms – pg. 48 Program.cs Searching Algorithms – pg. 49 N/A Performance Implications and Choosing Data Structures – pg. 50 N/A
Lesson 3 Video time - 00:15:48 Exercise Lab time - 00:28:00 Workbook time - 01:00:00	Version Control Systems GitHub Check Out Check In Branch Merge Clone	2.4 Describe the purpose of version control systems 2.4.1 GitHub 2.4.2 Check-in 2.4.3 Check-out 2.4.4 Merge 2.4.5 Branch 2.4.6 Rollback	Creating a Repository Editing Files Creating Branches Creating a Clone Conflicts	GitHub - pg. 53 N/A Check Out and Check In - pg. 54 N/A Branch - pg. 55 N/A Merge and Clone - pg. 56 N/A

Domain 2 - Software Development Principles [approximately 6 hours of videos, labs, and projects]

Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
	Rollback Resolve Conflicts Secure Coding Concepts Encryption, Hashing, and Digital Signatures Keys Mitigating CSRF SQL Injection Risks of Using IFrames	2.4.7 Clone 2.4.8 Resolving conflicts 2.5 Describe secure coding concepts 2.5.1 Encryption, hashing and digital signatures 2.5.2 Public, private, and shared keys 2.5.3 Mitigating cross-site request forgery (CSRF) 2.5.4 SQL injection 2.5.5 Risks of using IFrames	Securing Data CSRF Prevention	Rollback and Resolving Conflicts - pg. 57 N/A Encryption, Hashing, and Digital Signatures - pg. 58 N/A Keys - pg. 59 N/A Mitigating CSRF - pg. 60 N/A SQL Injection and IFrames - pg. 61 N/A
Post-Assessment Assessment time - 01:00:00	Software Development Principles: Post-Assessment			

Domain 3 Lesson Plan

Domain 3 - Object-Oriented Programming [approximately 3.5 hours of videos, labs, and projects]				
Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
Pre-Assessment Assessment time - 00:30:00	Object-Oriented Programming: Pre-Assessment			
Lesson 1 Video time - 00:13:01 Exercise Lab time - 00:08:00 Workbook time - 00:30:00	Classes Properties, Methods, Events, Fields, and Constructors Create Classes Use Classes in Code Access Modifiers Instantiation Static vs. Instance Encapsulation	3.1 Construct, analyze, and use classes 3.1.1 Properties, methods, events, fields, and constructors 3.1.2 How to create classes 3.1.3 How to use classes in code 3.1.4 Access modifiers 3.1.5 Instantiation 3.1.6 Static vs. instance 3.1.7 Encapsulation	Understanding Classes Access Modifiers	Properties, Methods, Events, Fields, and Constructors - pg. 64 N/A Creating and Using Classes - pg. 65 N/A Access Modifiers and Instantiation - pg. 66 N/A Static Classes and Encapsulation - pg. 67
Lesson 2 Video time - 00:11:40 Exercise Lab time - 00:08:00 Workbook time - 00:30:00	Inheritance Base Class to Derived Class Generic Classes Abstract Classes Polymorphism Extending Class Functionality Overriding Methods Interfaces Overloading	3.2 Construct and analyze code that uses inheritance 3.2.1 Inherit the functionality of a base class into a derived class 3.2.2 Generic classes 3.2.3 Abstract classes 3.3 Construct and analyze code that uses polymorphism 3.3.1 Extending the functionality of a class after inheriting from a base class 3.3.2 Overriding methods in the derived class 3.3.3 Interfaces 3.3.4 Overloading	Using Classes for New Classes Understanding Abstract Classes	Base vs. Derived Classes and Generic vs. Abstract Classes - pg. 70 N/A Extending Class Functionality and Overriding Methods - pg. 71 N/A Interfaces and Overloading - pg. 72 N/A
Post-Assessment Assessment time - 01:00:00	Object-Oriented Programming: Post-Assessment			

Domain 4 Lesson Plan

Domain 4 - Web Applications [approximately 4.5 hours of videos, labs, and projects]				
Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
Pre-Assessment Assessment time - 00:30:00	Web Applications: Pre-Assessment			
Lesson 1 Video time - 00:14:20 Exercise Lab time - 00:24:00 Workbook time - 00:50:00	Construct and Analyze Web Applications HTML5, CSS3, and JavaScript Browser Developer Tools HTTP Request or Response State Management Cookies Local and Session Storage Page Lifecycle Event Model Client-Side vs. Server-Side Programming	4.1 Construct and analyze web applications 4.1.1 HTML5, CSS3, and JavaScript ES6 4.1.2 Browser developer tools 4.1.3 HTTP request or response 4.1.4 State management 4.1.5 Cookies 4.1.6 Local and session storage 4.1.7 Page lifecycle 4.1.8 Event model 4.1.9 Client-side vs. server-side programming	Button Logic Code Changes Retrieving Cookies Invoking Objects HTTP Status XMLHttpRequest	HTML, CSS, and JavaScript - pg. 75 N/A Developer Tools - pg. 76 N/A HTTP Requests, Responses, and State Management - pg. 77 N/A Cookies, Local Storage, and Session Storage - pg. 78 N/A ASP.NET Lifecycle - pg. 79 N/A Event Models and Programming Types - pg. 80 N/A
Lesson 2 Video time - 00:09:55 Exercise Lab time - 00:20:00 Workbook time - 00:45:00	Configure Web Hosting Create Virtual Directories and Websites Publish Web Applications Web Server Role Configure Web Services Client Application Consumption Accessing Web Services JSON REST API OAuth XML Architectural Patterns MVC MVVM SPA	4.2 Describe and configure web hosting 4.2.1 Creating virtual directories and websites 4.2.2 Publishing web applications 4.2.3 Describe the role of the web server 4.3 Describe and configure web services 4.3.1 Web services that are consumed by client applications 4.3.2 Accessing web services from client applications 4.3.3 JSON 4.3.4 REST API 4.3.5 OAuth 4.3.6 XML 4.4 Describe and identify architectural patterns' 4.4.1 Model-view-controller (MVC) 4.4.2 Model-view-viewmodel (MVVM) 4.4.3 Single page application (SPA)	Create a Website Create Virtual Directory Code Usage Returning Strings Returning Arrays	Creating Websites - pg. 83 N/A Web Servers and Publishing Web Applications - pg. 84 N/A Web Services - pg. 85 N/A JSON and REST APIs - pg. 86 N/A OAuth and XML - pg. 87 N/A Architectural Patterns - pg. 88 N/A
Post-Assessment Assessment time - 01:00:00	Web Applications: Post-Assessment			

Domain 5 Lesson Plan

Domain 5 - Databases [approximately 6 hours of videos, labs, and projects]				
Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
Pre-Assessment Assessment time - 00:30:00	Databases: Pre-Assessment			
Lesson 1 Video time - 00:18:48 Exercise Lab time - 00:20:00 Workbook time - 00:50:00	Design and Normalize a Database Characteristics and Capabilities Database Design ERDs Normalization Concepts Index Concepts Constraints Primary Key Foreign Keys Relationships	5.1 Design and normalize a database 5.1.1 Characteristics and capabilities of database products 5.1.2 Database design 5.1.3 Entity Relationship Diagrams (ERDs) 5.1.4 Normalization concepts (to 3NF) 5.1.5 Indexes 5.1.6 Constraints 5.1.7 Primary key 5.1.8 Foreign keys 5.1.9 Relationships	Diagrams Constraints Understanding Primary Keys Creating Foreign Keys	Characteristics of Databases - pg. 91 N/A Database Design - pg. 92 N/A Normalization - pg. 93 514-normalization1.xlsx Indexes and Constraints - pg. 94 N/A Primary and Foreign Keys - pg. 95 N/A Relationships - pg. 96 N/A
Lesson 2 Video time - 00:17:37 Exercise Lab time - 00:12:00 Workbook time - 00:40:00	SQL Queries Stored Procedures Updating and Selecting Data DML vs. DDL Functions Triggers Cursors Joins Indexes	5.2 Construct, analyze, and optimize ANSI SQL queries 5.2.1 Creating and accessing stored procedures 5.2.2 Updating and selecting data 5.2.3 DML vs. DDL 5.2.4 Functions 5.2.5 Triggers 5.2.6 Cursors 5.2.7 Joins 5.2.8 Indexes	Understanding Functions Executing a JOIN Statement	Stored Procedures - pg. 99 N/A Database Language - pg. 100 N/A Functions and Triggers - pg. 101 N/A Cursors and Indexes - pg. 102 N/A Joins - pg. 103 N/A
Lesson 3 Video time - 00:16:46 Exercise Lab time - 00:36:00 Workbook time - 00:45:00	Manage Transactions Commit Rollback Save Concurrency Isolation Levels Lock Database Access Methods Entity Framework Connection Pools LINQ NoSQL Databases Document Databases Key-Value Databases	5.3 Manage transactions 5.3.1 Commit 5.3.2 Rollback 5.3.3 Save 5.3.4 Concurrency 5.3.5 Isolation levels 5.3.6 Lock 5.4 Describe database access methods 5.4.1 Entity Framework (code-first and database-first) 5.4.2 Connection pools 5.4.3 LINQ 5.5 Describe types of NoSQL databases 5.5.1 Document databases 5.5.2 Key-value databases	Understanding Transactions Further Understanding Transactions More Transactions Concurrent Transactions Recognizing Databases Further Recognizing Databases	Commit and Rollback - pg. 106 N/A Save and Concurrency - pg. 107 N/A Isolation Levels and Locks - pg. 108 N/A Entity Frameworks, Connection Pools, and LINQ - pg. 109 N/A Document and Key-Value Databases - pg. 110 N/A

Domain 5 - Databases [approximately 6 hours of videos, labs, and projects]

Lesson	Lesson Topic and Subtopics	Objectives	Exercise Labs	Workbook Projects and Files
Post-Assessment Assessment time - 01:00:00	Databases: Post-Assessment			